15 Microbiology of Endodontic Infections

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CHAPTER OUTLINE

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Apical periodontitis is essentially an inflammatory disease of microbial etiology primarily caused by infection of the root canal system.[217] Although chemical and physical factors can induce periradicular inflammation, a large body of scientific evidence indicates that endodontic infection is essential to the progression and perpetuation of the different forms of apical periodontitis.[18,91,123,273] Endodontic infection develops in root canals devoid of host defenses, as a consequence of either pulp necrosis (as a sequel to caries, trauma, periodontal disease, or invasive operative procedures) or pulp removal for treatment.

Microorganisms other than bacteria have been detected in association with endodontic infections, including fungi, archaea, and viruses.[54,194,254,261,291] However, given their high prevalence, dominance, organization, and pathogenicity, bacteria can be regarded as the major microorganisms implicated in the pathogenesis of apical periodontitis. In advanced stages of the endodontic infectious process, bacterial structures resembling biofilms are commonly observed adhering to the canal walls.[124,164,245] Consequently, apical periodontitis has been included in the roll of biofilm-related oral diseases.[256] Bacteria colonizing the root canal system enter in contact with the periradicular tissues via apical/lateral foramina or root perforations. As a consequence of the encounter between bacteria and host defenses, inflammatory changes take place in the periradicular tissues and give rise to the development of apical periodontitis. Depending on several bacterial and host-related factors, apical periodontitis can be symptomatic (acute) or asymptomatic (chronic).

The ultimate goal of endodontic treatment is either to prevent the development of apical periodontitis or, in cases where the disease is already present, to create appropriate conditions for periradicular tissue healing. The intent is to preserve the tooth and associated periradicular tissues in healthy conditions. Because apical periodontitis is an infectious disease, the rationale for endodontic treatment is to eradicate the occurring infection or prevent microorganisms from infecting or reinfecting the root canal or the periradicular tissues. The cardinal principle of any health care profession is the thorough understanding of disease etiology and pathogenesis, which provides a framework for effective treatment. In this context, understanding the microbiologic aspects of apical periodontitis is the basis for sound endodontic practice and sets the stage for an evidence-based therapeutic approach. This chapter focuses on diverse aspects of endodontic microbiology, including pathogenetic, taxonomic, morphologic, and ecologic issues.

Apical Periodontitis as an Infectious Disease

The first recorded observation of bacteria in the root canal dates back to the 17th century and the Dutch amateur microscope builder Antony van Leeuwenhoek (1632–1723). He reported that the root canals of a decayed tooth "were stuffed with a soft matter" and that "the whole stuff" seemed to be alive.[44] At that time, the role of Leeuwenhoek's "animalcules"

in disease causation was unsuspected. It took almost 200 years until his observation was confirmed and a cause-and-effect relationship between bacteria and apical periodontitis was suggested. This occurred specifically in 1894, when Willoughby Dayton Miller, an American dentist working at the laboratory of Robert Koch in Berlin, Germany, published a milestone study reporting on the association between bacteria and apical periodontitis after an analysis of samples collected from root canals.[119] By means of bacterioscopy of the canal samples, he found bacterial cells in the three basic morphologies known at the time: cocci, bacilli, and spirilla (or spirochetes) (Fig. 15.1). Morphologically, the endodontic microbiota was clearly different in the coronal, middle, and apical parts of the root canal. Spirochetes were found in high frequencies in abscessed cases, and a pathogenic role was suspected for these bacteria. Most of the bacteria Miller observed under light microscopy could not be cultivated using the technology available at that time. Those bacteria were conceivably anaerobic bacteria, which were only successfully cultivated about 50 to 100 years later with the advent of anaerobic culture techniques. However, it is now widely recognized that a large number of bacterial species living in diverse environments still remain to be cultivated by current technology,[6,154,155] and the root canal is no exception (discussed later in this chapter). Based on his findings, Miller raised the hypothesis that bacteria were the causative agents of apical periodontitis.

Approximately 70 years after Miller's classic study, his assumptions were confirmed by an elegant study from Kakehashi et al.[91] These authors investigated the response of exposed dental pulps to the oral cavity in conventional and germ-free rats. Histologic evaluation was performed and revealed that pulp necrosis and apical periodontitis lesions developed in all conventional rats; however, the exposed pulps of germ-free rats not only remained vital but also repaired themselves with hard-tissue formation. Dentin-like tissue sealed the exposure area and isolated the pulps again from the oral cavity.

The important role of bacteria in the etiology of apical periodontitis was further confirmed by Sundqvist's study.[273] This author applied advanced anaerobic culturing techniques to the evaluation of bacteria occurring in the root canals of teeth whose pulps became necrotic after trauma. Bacteria were found only in the root canals of teeth exhibiting radiographic evidence of apical periodontitis, confirming the infectious cause of this disease. Anaerobic bacteria accounted for more than 90% of the isolates. Findings from the Sundqvist's study also demonstrated that in the absence of infection, the necrotic pulp tissue itself and stagnant tissue fluid in the root canal do not induce or perpetuate apical periodontitis lesions.

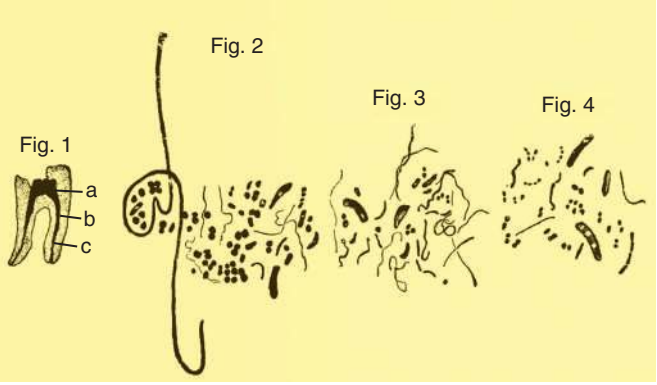


Fig. 15.1 Drawings from Miller's classic paper showing different bacterial forms in a root canal sample observed by microscopy.

Möller et al.[123] also provided strong evidence about the microbial causation of apical periodontitis. Their study on monkeys demonstrated that only devitalized pulps that were infected induced apical periodontitis lesions, whereas devitalized and noninfected pulps showed an absence of significant pathologic changes in the periradicular tissues. In addition to corroborating the importance of microorganisms for the development of apical periodontitis, this study also confirmed that necrotic pulp tissue per se is unable to induce and maintain an apical periodontitis lesion.

Bacteria causing apical periodontitis are primarily organized in biofilms colonizing the root canal system. Bacterial agglomerations resembling what is known nowadays as biofilms had been observed by several morphologic studies,[124,210,245] but it was not until the study by Ricucci and Siqueira[164] that the high prevalence of bacterial biofilms was consistently disclosed and a strong association with primary and posttreatment apical periodontitis was reported (see the section Spatial Distribution of the Endodontic Microbiota for a further discussion of biofilms in endodontic infections).

Routes of Root Canal Infection

Under normal conditions, the pulpodentin complex is sterile and isolated from the oral microbiota by overlying enamel and cementum. In the event that the integrity of these natural layers is breached (e.g., as a result of caries, trauma-induced fractures and cracks, restorative procedures, scaling and root planning, attrition, abrasion) or naturally absent (e.g., because of gaps in the cementum coating at the cervical root surface), the pulpodentin complex is exposed to the oral environment. The pulpodentin complex is then challenged by bacteria present in caries lesions, saliva bathing the exposed area, or dental plaque formed on the exposed area. Bacteria from subgingival biofilms associated with periodontal disease may also have access to the pulp via dentinal tubules at the cervical region of the tooth and lateral or apical foramina. Bacteria and other microorganisms may also have access to the root canal any time during or after endodontic intervention, emphasizing the need for maintaining strict asepsis during treatment and promoting an adequate coronal seal.

Whenever dentin is exposed, the pulp is put at risk of infection as a consequence of the permeability of normal dentin dictated by its tubular structure (Fig. 15.2).[140] Dentinal tubules traverse the entire width of the dentin and have a conical conformation, with the largest diameter located near the pulp (mean, 2.5 μm) and the smallest diameter near the enamel or cementum (mean, 0.9 μm).[58] The smallest tubule diameter is entirely compatible with the cell diameter of most oral bacterial species, which usually ranges from 0.2 to 0.7 μm. One might assume that once exposed, dentin offers an unimpeded pathway for bacteria

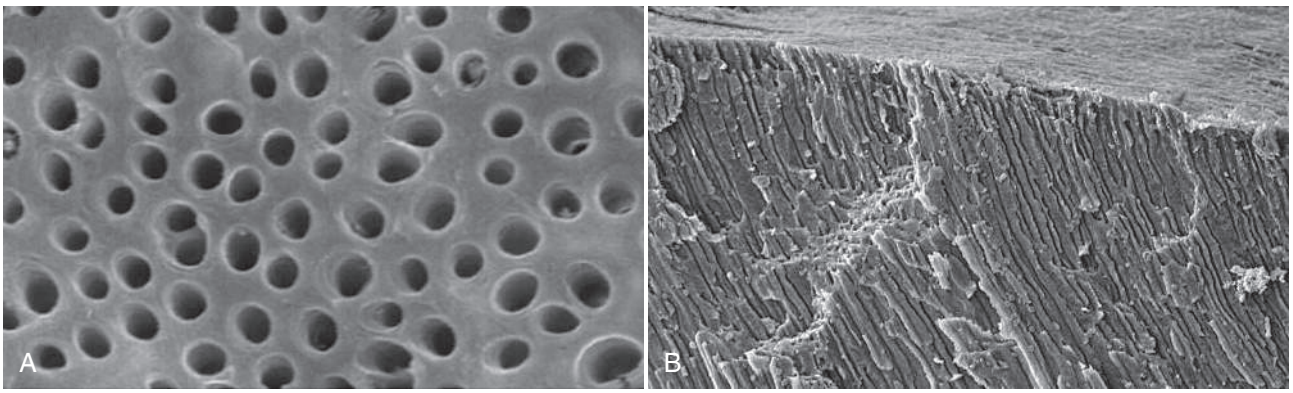


Fig. 15.2 **A,** Scanning electron micrographs of dentin showing tubules in cross-sectional view (magnification ×850). **B,** Longitudinal view (magnification ×130).

to reach the pulp via tubules. However, it has been demonstrated that bacterial invasion of dentinal tubules occurs more rapidly with a nonvital pulp than with a vital pulp.[128] With a vital pulp, outward movement of dentinal fluid and the tubular contents (including odontoblast processes, collagen fibrils, and the sheathlike lamina limitans that lines the tubules) influence dentinal permeability and can conceivably delay intratubular invasion by bacteria. Because of the presence of tubular contents, the functional or physiologic diameter of the tubules is only 5% to 10% of the anatomic diameter seen by microscopy.[117] Other factors such as dentinal sclerosis, tertiary dentin, smear layer, and intratubular deposition of fibrinogen also reduce dentin permeability and thereby limit or even impede bacterial progression to the pulp via dentinal tubules.[142] Host defense molecules, such as antibodies and components of the complement system, may also be present in the dentinal fluid of vital teeth and can assist in the protection against deep bacterial invasion of dentin.[3,133,134] As long as the pulp is vital, dentinal exposure does not represent a significant route of pulpal infection, except when dentin thickness is considerably reduced or when the dentin permeability is significantly increased.

Most of the bacteria in the carious process are nonmotile; they invade dentin by repeated cell division, which pushes cells into tubules. Bacterial cells may also be forced into tubules by hydrostatic pressures developed on dentin during mastication.[118] Bacteria inside tubules under a deep carious lesion can reach the pulp even before frank pulpal exposure.[85] As previously mentioned, it has been assumed that the pulp will not be easily infected if it is still vital. The few bacteria that reach the pulp may not be significant, because the vital pulp can eliminate such a transient infection and rapidly clear or remove bacterial products. This efficient clearance mechanism tends to prevent injurious agents from reaching a high enough concentration to induce significant inflammatory reactions.[141] On the other hand, if the vitality of the pulp is compromised and the defense mechanisms are impaired, even a small amount of bacteria may initiate pulpal infection.

Direct exposure of the dental pulp to the oral cavity is the most obvious route of endodontic infection. Caries is the most common cause of pulp exposure, but bacteria may also reach the pulp via direct pulp exposure as a result of iatrogenic restorative procedures or trauma. The exposed pulp tissue comes in direct contact with oral bacteria from carious lesions, saliva, or plaque accumulated onto the exposed surface. Almost invariably, exposed pulps will undergo inflammation and necrosis and become infected. The time elapsed between pulp exposure and infection of the entire canal is unpredictable, but it is usually a slow process.[35]

The egress of bacteria and their products from infected root canals through apical, lateral, or furcation foramina, dentinal tubules, and iatrogenic root perforations can directly affect the surrounding periodontal tissues and give rise to pathologic changes in these tissues. Conceptually, bacteria in subgingival plaque biofilms associated with periodontal disease could reach the pulp by the same pathways intracanal bacteria reach the periodontium and could thereby exert harmful effects on the pulp. It has been demonstrated that although degenerative and inflammatory changes of different degrees may occur in the pulp of teeth with associated periodontal disease, significant pulpal necrosis only develops if the periodontal biofilm reaches the apical foramen, leading to irreversible damage to the main blood vessels that penetrate through this foramen (Fig. 15.3).[102] After the pulp becomes necrotic, periodontal bacteria can invade the root canal system via apical foramina, lateral canals, and exposed dentinal tubules.

In the past, it was claimed that bacteria could reach the pulp by anachoresis.[72] Theoretically, bacteria can be transported in the blood or lymph to an area of tissue damage, where they leave the vessel, enter the damaged tissue, and establish an infection.[61,170] However, there is no clear evidence showing that this process represents a route for root canal infection. Bacteria could not be recovered from unfilled root canals of cats when the bloodstream was experimentally infected, unless the root canals were overinstrumented during the period of bacteremia, with resulting injury to periodontal blood vessels and blood seepage into the canal.[40] Another argument against anachoresis as a route for pulpal infection comes from the study by Möller et al.,[123] who induced pulpal necrosis in monkeys' teeth and reported that all cases of aseptic necrosis remained bacteria-free after 6 to 7 months of observation.

Bacteria have been isolated from traumatized teeth with necrotic pulps with apparently intact crowns.[273,306] Although anachoresis has been suggested to be the mechanism through which these traumatized teeth become infected,[72] current evidence indicates that the main pathway

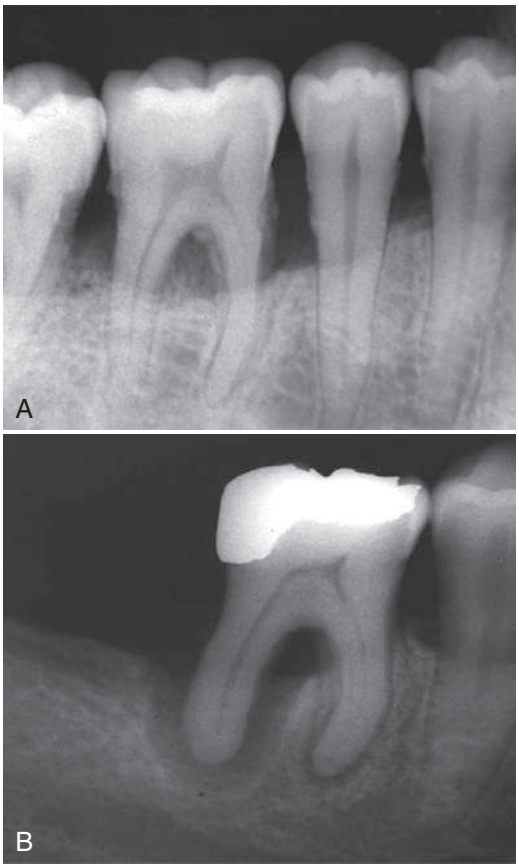


Fig. 15.3 Periodontal disease **(A)** mostly affects the pulp vitality when the subgingival biofilm reaches the apical foramen **(B)**.

of root canal infection in these cases is dentinal exposure due to enamel cracks.[111,112] Macro- and microcracks in enamel can occur in most teeth (not only traumatized teeth) and do not necessarily end at the enamel-dentin junction; they can extend deep into the dentin.[112] A large number of dentinal tubules are exposed to the oral environment by a single crack. Cracks can be clogged with plaque biofilm and provide portals of entry for bacteria. If the pulp remains vital after trauma, bacterial penetration into tubules is counteracted by the dentinal fluid and tubular contents, as discussed earlier, and pulpal health is not usually jeopardized. But if the pulp becomes necrotic as a consequence of trauma, it loses the ability to protect itself against bacterial invasion, and regardless of dentin thickness, dentinal tubules may become true avenues through which bacteria reach and colonize the necrotic pulp.

Whatever the route of bacterial access to the root canal, necrosis of pulp tissue is a prerequisite for the establishment of primary endodontic infections. To reiterate: If the pulp is vital, it can protect itself against bacterial invasion and colonization. If the pulp becomes necrotic as a sequel to caries, trauma, operative procedures, or periodontal disease, then it can be easily infected.

Another situation in which the root canal system is devoid of host defenses relates to cases in which the pulp was removed for treatment. Microbial penetration in the canal can occur during treatment, between appointments, or even after root canal obturation. The main causes of microbial introduction into the canal *during treatment* include residual dental plaque biofilm, calculus, or caries on the tooth crown; leaking rubber dam; contaminated endodontic instruments (e.g., after touching with the fingers); and contaminated irrigant solutions or other solutions of intracanal use (e.g., saline solution, distilled water, citric acid). Bacteria can also enter the root canal system *between appointments* by leakage through the temporary restorative material; breakdown, fracture, or loss of the temporary restoration; fracture of the tooth structure; and in teeth left open for drainage. Bacteria can penetrate the root canal system even *after completion of the root canal obturation* by leakage through the temporary or permanent restorative material; breakdown, fracture, or loss of the temporary/permanent restoration; fracture of the tooth structure; recurrent decay; or delay in the placement of the permanent restoration.[227]

Mechanisms of Microbial Pathogenicity and Virulence Factors

Pathogenicity is a term that refers to the ability of a microorganism to cause disease. The degree of pathogenicity of a microorganism is referred to as *virulence*. Microbial secreted products, structural cellular components, and strategies that contribute to pathogenicity are regarded as *virulence factors*. One example of bacterial strategy that contributes to pathogenicity is the ability to form biofilms, which confers protection against host defenses, microbial competitors, and antimicrobial agents. Some microorganisms routinely cause disease in a given host and are called *primary pathogens*. Other microorganisms cause disease only when host defenses are impaired and are called *opportunistic pathogens*. Bacteria that make up the normal resident microbiota are usually present as harmless commensals and live in balance with the host. One of the greatest beneficial effects of human microbiota is probably the tendency to protect the host from exogenous infections by excluding other microorganisms. Nevertheless, in certain situations, the balance may be disturbed by a decrease in the normal level of resistance, and then the commensal bacteria are usually the first to take advantage. Most bacteria involved with endodontic infections are normal inhabitants of the oral microbiota that exploit changes in the balance of the host–bacteria relationship, becoming opportunistic pathogens.

The bacterial species involved with the pathogenesis of primary apical periodontitis may have participated in the early stages of pulp inflammation and necrosis, or they may have gained entry into the root canal space any time after pulpal necrosis. In the former situation, involved bacteria are usually those present in the advanced front of caries lesions and from saliva bathing the affected area. Bacteria in caries lesions form authentic biofilms adhered to dentin (Fig. 15.4). Diffusion of bacterial products through dentinal tubules induces pulp inflammation long before the tissue is exposed. After frank exposure, the surface of the pulp tissue can be colonized and covered by bacteria present in the caries biofilm. The exposed pulp tissue enters in direct contact with bacteria and their products and responds with

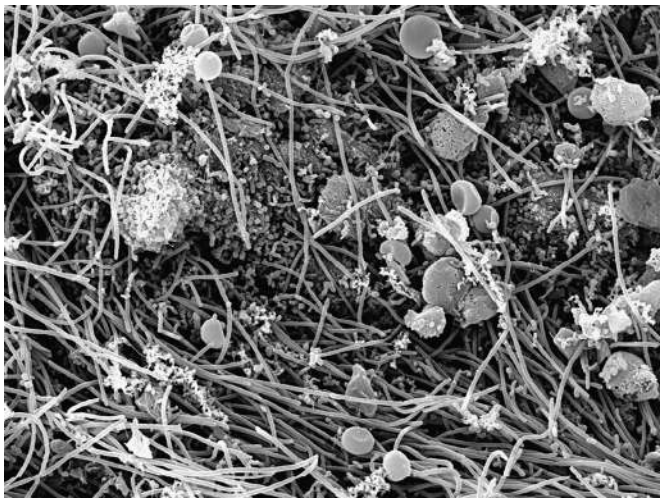


Fig. 15.4 Scanning electron micrograph showing a bacterial biofilm covering dentin in a deep carious lesion. Note the presence of different bacterial morphotypes (magnification ×3500). (From Torabinejad M, Walton RE: *Endodontics: principles and practice,* ed 5, St. Louis, 2015, Saunders/Elsevier.)

severe inflammation. Some tissue invasion by the involved bacteria may occur. Bacteria in the forefront of infection have to survive the attack from the host defenses and at the same time acquire nutrients to keep themselves alive. In this bacteria–pulp clash, the pulp invariably is "defeated" and becomes necrotic, so bacteria move forward and "occupy the territory"—that is, they colonize the necrotic tissue. These events advance through tissue compartments, coalesce, and move toward to the apical part of the canal until virtually the entire root canal is necrotic and infected (Fig. 15.5). At this stage, involved bacteria can be regarded as the early root canal colonizers or pioneer species.

Early colonizers play an important role in the initiation of the apical periodontitis disease process. Moreover, they may significantly modify the environment, making it conducive to the establishment of other bacterial groups. These new species may have access to the canal via coronal exposure or exposed dentinal tubules, establish themselves, and contribute to a shift in the microbiota. Rearrangement in the proportions of the pioneer species and latecomers occurs, and, as the environment changes, some early colonizers may even disappear. With the passage of time, the endodontic microbiota becomes more and more structurally and spatially organized.

Some virulence attributes required for pathogens to thrive in other sites may be of no value for bacteria that reach the root canal after necrosis—for instance, the ability to evade host defenses. This is because latecomers face no significant opposition from host defenses, which are no longer active in the canal after necrosis. Although colonization may appear an easy task for late colonizers, other environmental factors (e.g., interaction with pioneer species, oxygen tension, nutrient availability) will determine whether new species entering the canal will succeed in establishing themselves and join the early colonizers to make up a dynamic mixed community in the root canal. Ultimately, the root canals of teeth with radiographically detectable apical periodontitis lesions harbor both early colonizers that managed to stay in the canals and late colonizers that managed to adapt to the new environmental conditions.[234]

Periradicular inflammation can be observed, even before the frontline of intracanal bacterial infection reaches the apical foramen.[8,124,265,307] Bacteria exert their pathogenicity by wreaking havoc on the host tissues through direct or indirect mechanisms. Bacterial virulence factors that cause direct tissue harm include those that damage host cells or the intercellular matrix of the connective tissue. Secreted bacterial products, such as enzymes, exotoxins, heat-shock proteins, and metabolic end products, are the main examples.[234] Furthermore, bacterial structural components, including lipopolysaccharide (LPS), peptidoglycan, lipoteichoic acid (LTA), fimbriae, flagella, outer membrane proteins

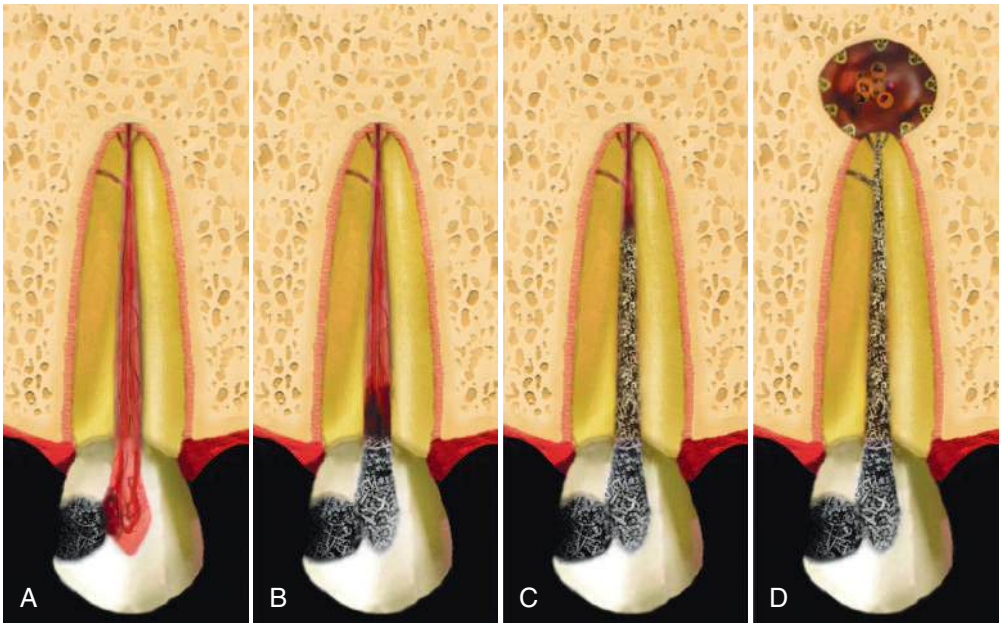


Fig. 15.5 Dynamics of pulp response from caries exposure **(A)** to pulp inflammation **(B)** to pulp necrosis **(C)** to apical periodontitis formation **(D)**.

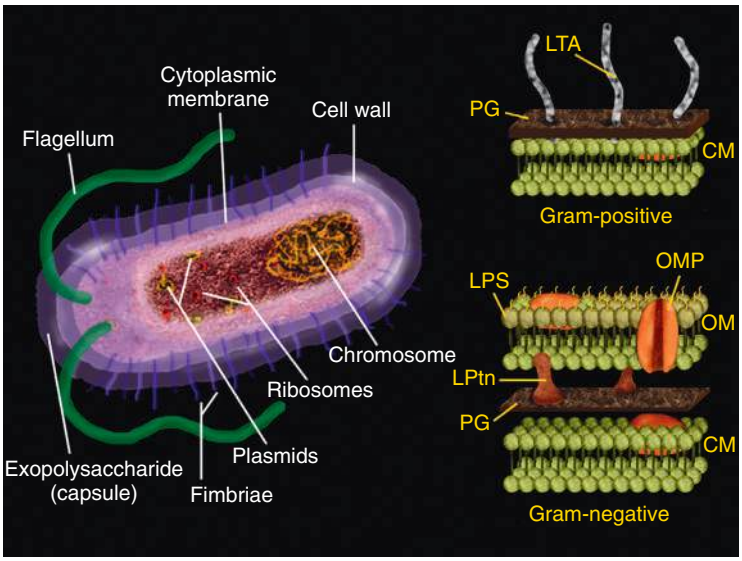


Fig. 15.6 Bacterial cell and its structural components that can act as virulence factors. *Right:* A detailed scheme of the bacterial cell walls from gram-positive and gram-negative bacteria. *CM,* Cytoplasmic membrane; *LPS,* lipopolysaccharide (endotoxin); *LPtn,* lipoproteins; *LTA,* lipoteichoic acid; *OM,* outer membrane; *OMP,* outer membrane protein; *PG,* peptidoglycan.

and vesicles, lipoproteins, DNA, and exopolysaccharides, can act as modulins by stimulating the development of host immune reactions capable not only of defending the host against infection but also of causing severe tissue destruction (Fig. 15.6).[80,234,287] For instance, inflammatory and noninflammatory host cells are stimulated by bacterial components to release proinflammatory cytokines that are involved in bone resorption characteristically observed in apical periodontitis.[201] Another example of indirect damage is pus formation in acute apical abscesses, which results from the destruction of the connective extracellular matrix by oxygen-derived free radicals and lysosomal enzymes released by polymorphonuclear leukocytes in response to bacteria.[285] Although direct damage caused by bacterial products may certainly be involved in the pathogenesis of apical periodontitis, bacterial indirect destructive effects seem to be more significant in this regard.[201]

Apical periodontitis is a multifactorial disease that is resultant of the interplay of many host and bacterial factors. Few if any of the putative endodontic pathogens are individually capable of inducing all of the events involved in the pathogenesis of the different forms of apical periodontitis. Therefore, the process requires an integrated and orchestrated interaction of the selected members of the mixed endodontic microbiota and their respective virulence attributes.[295] LPS, which is exclusive of gram-negative bacteria, has been detected in high concentrations in canals of teeth with large and/or symptomatic apical periodontitis, and in conditions of persistent exudation.[38,64,84,88,205,206] Although LPS has been widely studied and quoted as an important virulence factor with regard to the pathogenesis of apical periodontitis,[39,127] many other bacterial mediators are also expected to contribute to disease causation. For instance, some cases of primary infections and many cases of secondary/persistent infections may harbor exclusively gram-positive bacteria, which indicates that the involvement of other factors must not be overlooked. Indeed, LTA, which is exclusive of gram-positive bacteria, has been detected in all teeth with posttreatment apical periodontitis.[10] Many bacterial secreted products have been detected in endodontic infections.[79,152,153] Therefore, the pathogenesis of different forms of apical periodontitis and even the same form in different individuals is unlikely to follow a stereotyped course with regard to the bacterial mediators involved.[218]

Spatial Distribution of the Endodontic Microbiota

Mounting evidence indicates that apical periodontitis, like caries and periodontal diseases, is also a biofilm-related disease. Morphologic studies have shown that the root canal microbiota in primary infections is dominated by bacterial morphotypes that include cocci, rods, filaments, and spirilla (spirochetes) (Fig. 15.7). Fungal cells are sporadically found (Fig. 15.8).[210,245] Although planktonic bacterial cells suspended in a fluid phase and enmeshed in necrotic pulp tissue can be observed in the main root canal, most bacteria colonizing the root canal system usually grow in sessile multispecies biofilms adhered to the dentinal walls (Fig. 15.9).[124,165,245] Lateral canals, apical ramifications, and isthmuses connecting main canals may also be clogged with bacterial biofilms (Figs. 15.10 and 15.11).[160,163,290]

Bacterial cells at the bottom of the endodontic biofilm structure are often seen penetrating the dentinal tubules (Fig. 15.12). Dentinal tubule infection occurs in 70% to 80% of the teeth evincing apical periodontitis lesions.[116,148] A shallow penetration is more common, but in some teeth bacterial cells are observed reaching approximately 300 μm deep (Fig. 15.13).[245] Dividing bacterial cells are frequently seen within tubules (see Fig. 15.13),[245] indicating that they are alive and probably derive nutrients from degrading odontoblastic processes, denatured collagen, bacterial cells that die during the course of infection, and intracanal fluids that enter the tubules by capillarity.

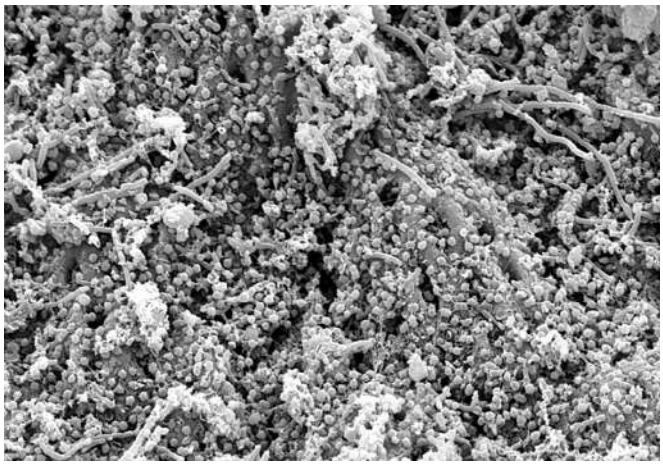


Fig. 15.7 Mixed bacterial population colonizing the root canal wall. Cocci are the predominant forms, but rods, filaments, and spirochetes are also observed. In some areas, coccoid cells are relatively apart from each other (magnification ×2200). (From Siqueira JF Jr, Rôças IN, Lopes HP: Patterns of microbial colonization in primary root canal infections. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:174, 2002.)

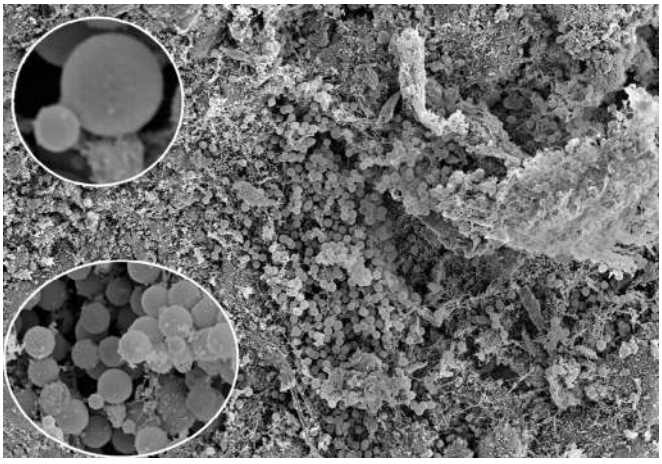


Fig. 15.8 Heavy colonization of yeast cells in the root canal of an extracted tooth with primary infection associated with apical periodontitis (magnification ×300). Note that some cells are in the stage of budding. A daughter cell is growing on the surface of the mother cell (insets: magnification ×2700 [*bottom*], magnification ×3500[*top*]). (From Siqueira JF Jr, Sen BH: Fungi in endodontic infections. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 97:632, 2004.)

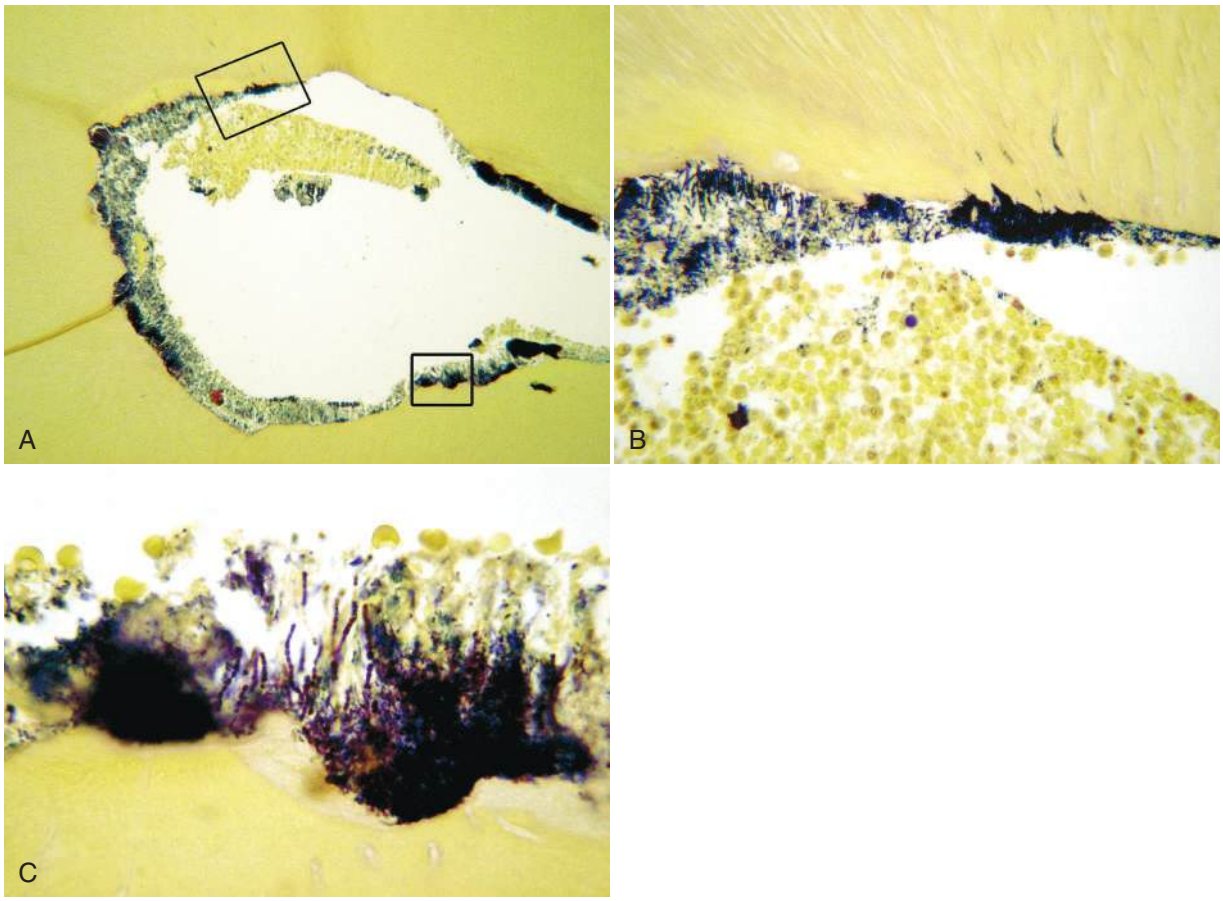


Fig. 15.9 A, Biofilm on the walls of a mesial root canal from a mandibular first molar (magnification ×100). The tooth was symptomatic, and an apical periodontitis lesion was present. Sections in **B** and **C** correspond to higher magnifications of the larger and smaller insets, respectively. Note the accumulation of polymorphonuclear neutrophils in the canal near the biofilm. (**B,** magnification ×400; **C,** magnification ×1000). Sections stained with a Taylor-modified Brown and Brenn technique. (Courtesy Dr. Domenico Ricucci.)

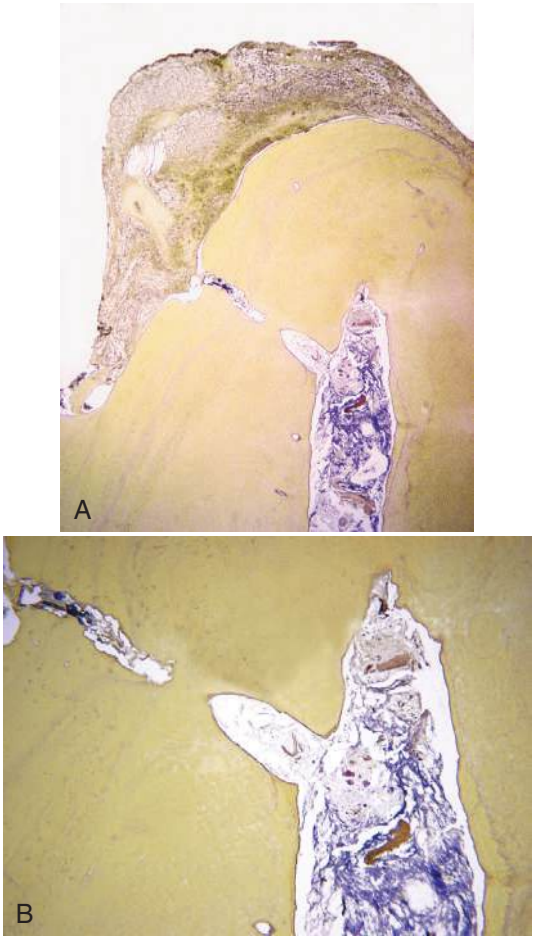


Fig. 15.10 A, Bacterial biofilm in the necrotic canal and in an apical ramification contiguous to inflamed periradicular tissues (magnification ×25). **B,** Higher magnification of **A** (magnification ×100). Sections stained with a Taylor-modified Brown and Brenn technique. (Courtesy Dr. Domenico Ricucci.)

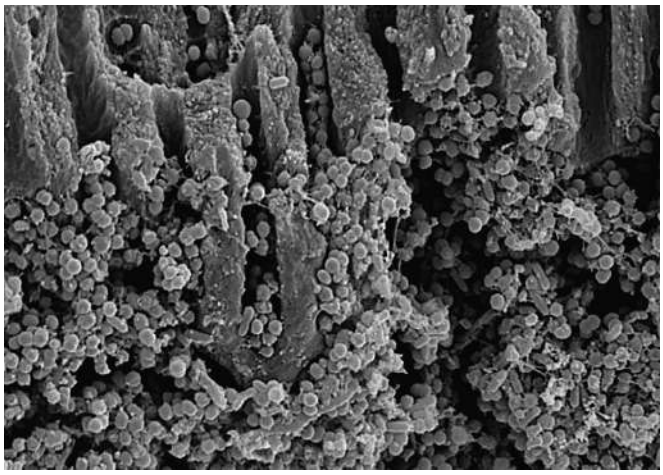


Fig. 15.12 Heavy infection of the root canal walls, mainly by cocci, but some small rods are also seen. Cocci are penetrating into dentinal tubules (magnification ×3500). (From Siqueira JF Jr, Rôças IN, Lopes HP: Patterns of microbial colonization in primary root canal infections. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:174, 2002.)

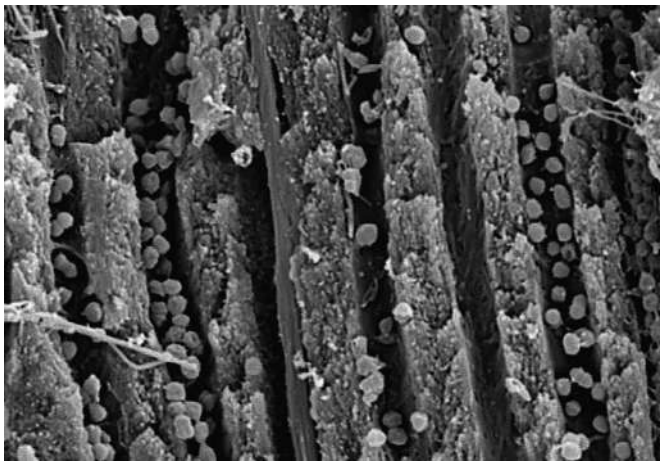


Fig. 15.13 Cocci in dentinal tubules approximately 300 μm from the main root canal (magnification ×5000). (From Siqueira JF Jr, Rôças IN, Lopes HP: Patterns of microbial colonization in primary root canal infections. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:174, 2002.)

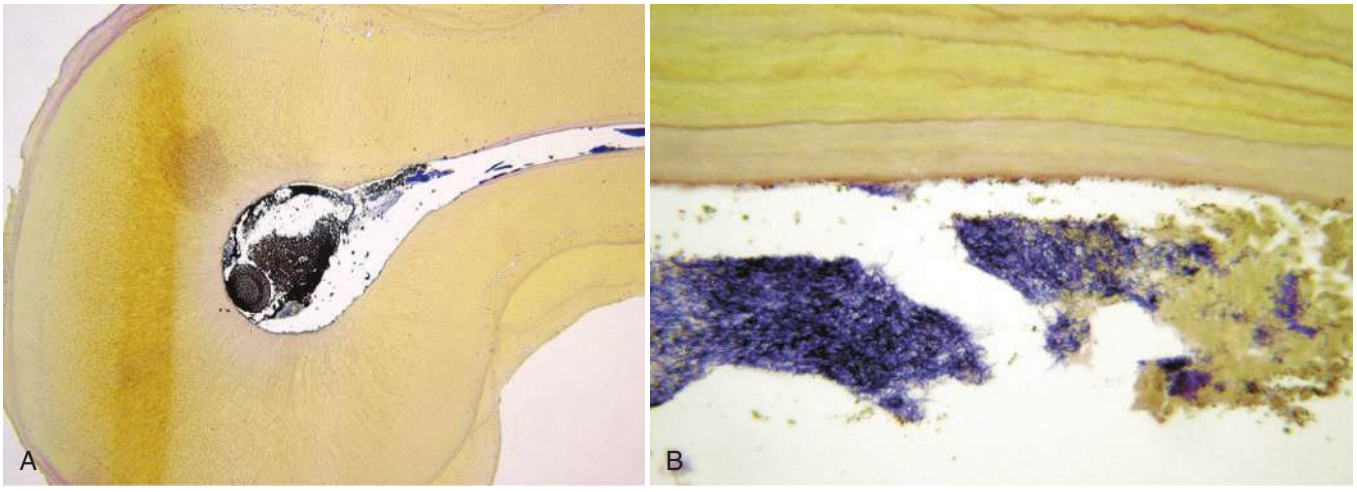


Fig. 15.11 Transverse sections of a maxillary second molar with fused mesial and palatal roots. **A,** Heavy bacterial infection of the canal, spreading to an isthmus (magnification ×25). **B,** Higher magnification of the isthmus clogged with bacteria (magnification ×400). Sections stained with a Taylor-modified Brown and Brenn technique. (Courtesy Dr. Domenico Ricucci.)

Several putative endodontic pathogens have been shown to be capable of penetrating dentinal tubules in vitro, including *Porphyromonas endodontalis*, *Porphyromonas gingivalis*, *Fusobacterium nucleatum*, *Actinomyces israelii*, *Propionibacterium acnes*, *Enterococcus faecalis*, *Candida albicans*, and streptococci.[112,146,223,246,299] A clinical study isolated and identified bacteria present in root dentin at different depths, and the most common isolates belonged to the genera *Prevotella, Porphyromonas, Fusobacterium, Veillonella, Peptostreptococcus, Eubacterium, Actinomyces,* lactobacilli, and streptococci.[148] Using immunohistologic analysis, Matsuo et al.[116] observed the occurrence of *F. nucleatum*, *Pseudoramibacter alactolyticus*, *Eubacterium nodatum*, *Lactobacillus casei*, and *Parvimonas micra* inside dentinal tubules from the canal walls of extracted teeth with apical periodontitis.

Whereas bacteria present as planktonic cells in the main root canal may be easily accessed and eliminated by instruments and antimicrobial substances used during endodontic treatment, those organized in biofilms attached to the canal walls or located into isthmuses, recesses, lateral canals, and dentinal tubules are definitely more difficult to reach,[290] and may require special therapeutic strategies to be eradicated.

Biofilm and Community-Based Microbial Pathogenesis

Individual microorganisms proliferating in a habitat give rise to *populations*. Such populations often occur as microcolonies in the environment. Populations interact with one another to form a *community*. The community and habitat are part of a larger system called an *ecosystem,* which is defined as a functional self-supporting system that includes the microbial community and its environment.

It has been recognized that the dental plaque biofilms associated with caries and periodontal diseases represent sophisticated communities that exert functions essential for the biofilm architecture and physiology, with consequent pathogenetic implications. Evidence indicates that apical periodontitis can also develop as a result of collaborative activities of a biofilm community established in the root canal system.

For many endogenous infections in the human body, the community, instead of a single species, has been regarded as the unit of pathogenicity. This community-as-pathogen concept is based on a holistic approach to understand the group behavior of bacteria in mixed infections, and assumes that the whole is more than the simple sum of its parts. Factors such as the composition (types of species), abundance (proportion of each species), interactions (synergism), and virulence factors will act in concert to define the collective pathogenicity of a certain community and determine the disease outcome.

Community profiling studies revealed that bacterial composition of the endodontic microbiota differs consistently between individuals suffering from the same disease (Fig. 15.14).[30,113,184,251] This indicates that apical periodontitis has a heterogeneous etiology, where multiple bacterial combinations can play a role in disease causation. Interindividual variability is even more pronounced when different geographic locations are studied.[60,113,184,244] Moreover, community structure differs significantly between

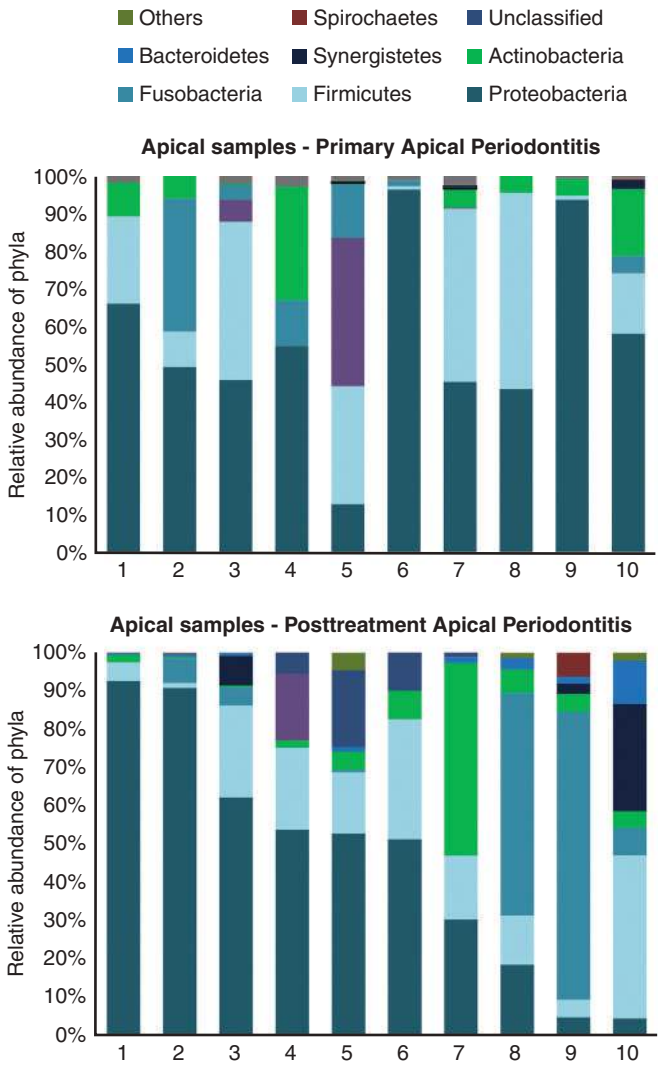


Fig. 15.14 High interindividual variability in bacterial diversity (richness and abundance) at the phylum level in samples taken from cryopulverized apices of teeth with primary and posttreatment apical periodontitis. Variability is much more pronounced at the genus and species (data not shown). (Based on data from Siqueira JF Jr, Alves FR, Rôças IN. Pyrosequencing analysis of the apical root canal microbiota. *J Endod* 37:1499, 2011, and Siqueira JF Jr, Antunes HS, Rôças IN, Rachid CT, Alves FR. Microbiome in the apical root canal system of teeth with posttreatment apical periodontitis. *PLoS One* 11:e0162887, 2016).

different disease forms (e.g., asymptomatic apical periodontitis vs. acute apical abscess),[196,200,251] suggesting existence of a pattern associated with each form.

BIOFILM AND BACTERIAL INTERACTIONS

The community-forming ability can be regarded as essential for microbial survival in virtually all environments. Indeed, the majority of microorganisms in nature invariably grow and function as members of metabolically integrated communities, or biofilms.[33,115] *Biofilm* can be defined as a sessile multicellular microbial community characterized by cells that are firmly attached to a surface and enmeshed in a self-produced matrix of extracellular polymeric substance (EPS), mostly composed of polysaccharide but also proteins and nucleic acids (Fig. 15.15).[33,45,75] The ability to form

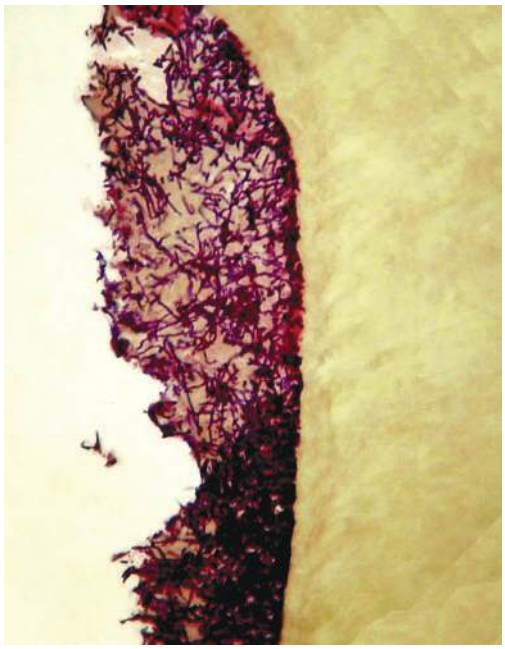


Fig. 15.15 Mixed bacterial biofilm adhered to the tooth surface (Brown and Brenn staining, magnification ×1000).

biofilms has been regarded as a virulence factor,[75] and biofilm infections account for an estimated 65% to 80% of bacterial infections that affect humans in the developed world.[32] Given its importance in varied aspects, biofilm properties have been extensively studied not only in medical microbiology but also in different sectors of industrial and environmental microbiology.

As a community, biofilms are composed of populations or microcolonies (±15% by volume) that are embedded and nonrandomly distributed in the EPS matrix (±85% by volume) and separated by water channels.[34,45,263,267] Dental biofilms can reach up to 300 or more cell layers in thickness.[263] Individual microcolonies may consist of a single bacterial species, but more frequently are composed of several different species in a mixed community.

The EPS matrix is not only important physically as part of the scaffold that determines the biofilm structure, but it is also biologically active and can retain nutrients, water, and essential enzymes within the biofilm.[5] The matrix also protects the biofilm community from exogenous threats and may participate in adherence to the surfaces.

In most biofilms in nature, bacterial populations arise from surface colonization by planktonic (unattached) bacterial cells. Floating bacteria approach to the surface to be colonized and attach to host proteins and/or coaggregate with previously bound bacteria. This process may occur in canals that are frankly exposed to the oral cavity and filled with saliva. However, biofilm formation in the root canal system in most circumstances, especially as a sequel to caries, is expected to occur in a different way. As the caries biofilm reaches the pulp, the latter becomes inflamed and necrotic, and the biofilm advances in apical direction. Therefore, biofilm formation on the root canal walls in most cases occurs gradually as the pulp becomes necrotic and the infectious process advances towards the apical canal.[256]

Bacterial interactions in biofilms lead to changes in growth rate, gene expression, and protein production. Genes expressed by cells in biofilms can differ by 20% to 70% from those expressed by the same cells growing in planktonic culture.[17,135,203] Thus, bacteria in biofilms adopt a radically different phenotype compared with their planktonic counterparts. Within biofilms, some bacteria also use sophisticated systems of cell-cell communication (quorum sensing) to coordinate gene expression. Phenotypic heterogeneity in biofilms is also observed as a result of exposure of microcolonies to a variety of gradients (e.g., oxygen tension, pH, type and amount of nutrients, cell density), which contribute to form diverse microenvironments throughout the biofilm structure.

BIOFILM RESISTANCE TO ANTIMICROBIAL AGENTS

From a clinical standpoint, the biofilm increased resistance to antimicrobial agents is of special concern. Bacterial cells arranged in biofilms are considered more resistant to antibiotics than the same cells grown in planktonic state. The antibiotic concentration required to kill bacteria in the biofilm is about 100 to 1000 times higher than that needed to kill the same species in planktonic state.[114]

Several possible mechanisms help explain the biofilm resistance to antimicrobials: (a) the biofilm structure may restrict penetration of antimicrobial agents[266]; (b) starved bacteria entering the stationary phase in biofilms are much less susceptible to antimicrobials than rapidly-growing cells[75]; and (c) presence of a subpopulation of specialized survivor cells known as persisters.[94]

Apical Periodontitis as a Biofilm-Related Disease

Numerous morphologic studies have reported the occurrence of bacterial agglomerations on the root canal walls resembling structures that later were defined as biofilm.[23,124,163,166,167,204,245] However, the strongest evidence about the association of bacterial biofilms with apical periodontitis came from a histobacteriologic study by Ricucci and Siqueira.[164] These authors evaluated the prevalence of biofilms in the apical root canal system of untreated teeth with primary apical periodontitis and treated teeth with posttreatment disease and looked for associations between biofilms and clinical/histopathologic conditions. Some of the most important findings of their study are as follows:

1. Intraradicular biofilms were generally observed in the apical segment of approximately 80% of the root canals of teeth with primary or posttreatment apical periodontitis.
2. Morphology of endodontic biofilms differed consistently from individual to individual (e.g., thickness, morphotypes, bacterial cells/extracellular matrix ratio).
3. Dentinal tubules underneath biofilms were often invaded by bacterial cells from the bottom of the biofilm community.
4. Biofilms were also commonly seen covering the walls of apical ramifications, lateral canals, and isthmuses.
5. Bacterial biofilms were more frequent in root canals of teeth with large apical periodontitis lesions. Because it

takes time for apical periodontitis to develop and become radiographically visible, one can surmise that large lesions represent a longstanding pathologic process caused by an even "older" intraradicular infection. In longstanding infectious processes, involved bacteria may have had enough time and conditions to adapt themselves to the environment and set a mature and organized biofilm community. The fact that the apical root canal of teeth with large lesions harbors a large number of bacterial cells and species almost always organized in biofilms may help explain the concept that treatment outcome is influenced by lesion size.[31,131]

6. The prevalences of intraradicular biofilms in teeth associated with apical cysts, abscesses, and granulomas were 95%, 83%, and 69.5%, respectively. Biofilms were significantly associated with epithelialized lesions. Because apical cysts develop as a result of epithelial proliferation in some granulomas,[107] it may be anticipated that the older the apical periodontitis lesion, the greater the probability of it becoming a cyst. Similar to teeth with large lesions, the age of the pathologic process may also help explain the high prevalence of biofilms in association with cysts.
7. Extraradicular biofilms were infrequent; they occurred in only 6% of the cases. Except for one case, they were always associated with intraradicular biofilms. All cases showing an extraradicular biofilm exhibited clinical symptoms.
8. Bacteria were also seen in the lumen of the main canal, ramifications, and isthmuses as flocs and planktonic cells, either intermixed with necrotic pulp tissue or possibly suspended in a fluid phase. Bacterial flocs are sometimes regarded as "planktonic biofilms" and may originate from growth of cell aggregates/coaggregates in a fluid or they may have detached from biofilms.[76]

Ricucci and Siqueira[164] also used the following criteria available in the literature to establish a causal link between biofilms and a given infectious disease[76,139,164]:

1. Infecting bacteria are adhered to or associated with a surface.
2. Direct examination of the infected tissue shows bacteria forming clusters or microcolonies encased in an extracellular matrix.
3. The infection is generally confined to a particular site and although dissemination may occur, it is a secondary event.
4. The infection is difficult or impossible to eradicate with antibiotics in spite of the responsible microorganisms being susceptible to killing in the planktonic cell state.
5. Ineffective host clearance is evident, as suggested by the location of bacterial colonies in areas of the host tissue associated with host inflammatory cells. Accumulation of polymorphonuclear neutrophils and macrophages surrounding bacterial aggregates/coaggregates in situ considerably increases the suspicion of biofilm involvement with disease causation.
6. Elimination or significant disruption of the biofilm structure and ecology leads to remission of the disease process.

Based on findings from the Ricucci and Siqueira's study,[164] apical periodontitis can be considered to fulfill four of the six criteria. Bacterial agreggates/coaggregates are observed adhered to or at least associated with the dentinal root canal walls (criterion 1). Bacterial colonies are often seen encased in an amorphous extracellular matrix (criterion 2). Endodontic biofilms are frequently confined to the root canal system, in only a few cases extending to the external root surface, but dissemination through the lesion never occurred (criterion 3). In the great majority of cases, biofilms are directly faced by an accumulation of inflammatory cells, especially polymorphonuclear neutrophils (criterion 5).

As for criterion 4, it is widely known that intraradicular endodontic infections cannot be effectively treated by systemic antibiotic therapy, even though most endodontic bacteria in the planktonic cell state are susceptible to currently used antibiotics.[15,65,96] The lack of efficacy of systemic antibiotics against intraradicular infections is mainly because the drug does not reach endodontic bacteria that are located in the avascular pulp necrotic space. The recognition of biofilms as the main mode of bacterial establishment in the root canal system further strengthens the explanations for the lack of antibiotic effectiveness against endodontic infections. Finally, there is a clear potential for fulfillment of criterion 6, because biofilms are frequently observed in canals of treated teeth with posttreatment apical periodontitis,[167] whereas teeth with a successful outcome show no biofilm infection of the root canal.[158] In addition, canals that exhibit negative cultures at the time of filling have a better treatment outcome.[52,296]

Methods for Microbial Identification

The endodontic microbiota has been traditionally investigated by culture methods. Culture is the process of propagating microorganisms in the laboratory by providing them with the required nutrients and proper physicochemical conditions, including temperature, moisture, atmosphere, salt concentration, and pH.[260] Essentially, culture analyses involve the following steps: sample collection and transport, dispersion, dilution, cultivation, isolation, and identification.[274] Endodontic samples are collected and transported to the laboratory in a viability-preserving, nonsupportive, anaerobic medium. They are then dispersed by sonication or vortex mixing, diluted, distributed onto various types of agar media, and cultivated under aerobic or anaerobic conditions. After a suitable period of incubation, individual colonies are subcultivated and identified on the basis of multiple phenotype-based aspects, including colony and cellular morphology, gram-staining pattern, oxygen tolerance, comprehensive biochemical characterization, and metabolic end-product analysis by gas-liquid chromatography. The outer cellular membrane protein profile is examined by gel electrophoresis, fluorescence under ultraviolet light, and susceptibility tests to selected antibiotics may be needed for identification of some species.[50] Marketed packaged kits that test for preformed enzymes have also been used for rapid identification of several species. Isolated colonies can also be identified by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS)[303] or 16S rRNA sequencing.[250]

Culture analyses of endodontic infections have provided a substantial body of information about the etiology of apical

periodontitis, composition of the endodontic microbiota in different clinical conditions, effects of treatment procedures in bacterial elimination, and susceptibilities of endodontic bacteria to antibiotics. Advantages and limitations of culture methods are listed in Box 15.1; some important limitations of culture methods make a comprehensive analysis of the endodontic microbiota difficult to achieve.

The difficulties in culturing or identifying many microbial species are of special concern. Unfortunately, not all microorganisms can be cultivated under artificial conditions, and this is simply because the nutritional and physiologic needs of most microorganisms are still unknown. Investigations of many environments using culture-independent methods have revealed that the large majority of microbial members of these systems still remain to be cultivated.[6,302] Furthermore, 40% to 80% of bacterial species composing the microbiota associated with diverse human sites, including the oral cavity, represent unknown and still uncultivated bacteria.[1,2,42,47,99,143,151,268]

That a given species has not been cultivated does not imply that this species will remain indefinitely impossible to cultivate. Myriad obligate anaerobic bacteria were uncultivable in the early 1900s, but further developments in anaerobic culturing techniques have to a large extent helped to solve this problem. It must be assumed that no single method or culture medium is suitable for isolating the vast diversity of microorganisms present in most environments.[69] Because we remain relatively unaware of the requirements for many bacteria to grow, identification methods not based on cultivability are required.

In some situations, even the successful cultivation of a given microorganism does not necessarily mean that this microorganism can be successfully identified. Culture-dependent identification is based on phenotypic traits observed in reference strains, with predictable biochemical and physical properties under optimal growth conditions. However, many phenotype-related factors can lead to difficulties in identification and even to misidentification.[16,20,240,282] As a consequence, phenotype-based identification does not always allow an unequivocal identification.

To sidestep the limitations of culturing, tools and procedures based on molecular biology have become available and have substantially improved the ability to achieve a more realistic description of the microbial world without the need for cultivation (Fig. 15.16). Molecular technology has also been applied to reliably identify cultivated bacteria, including strains with ambiguous or aberrant phenotypic behavior, rare isolates, poorly described or uncharacterized

Box 15.1 Advantages and Limitations of Culture Methods

Advantages	Limitations
Broad-range nature, identification of unexpected species	Impossibility of culturing a large number of extant bacterial species
Allow quantification of all major viable cultivable microorganisms in samples	Not all viable bacteria can be recovered
Allow determination of antimicrobial susceptibilities of isolates	Once isolated, bacteria require identification using a number of techniques
Physiologic studies are possible	Misidentification of strains with ambiguous or aberrant phenotypic behavior
Pathogenicity studies are possible	Low sensitivity
Widely available	Strict dependence on the mode of sample transport
	Samples require immediate processing
	Costly, time consuming, and laborious, as for cultivation of anaerobes
	Specificity is dependent on experience of microbiologist
	Extensive expertise and specialized equipment needed to isolate anaerobes
	Takes several days to weeks to identify most anaerobes

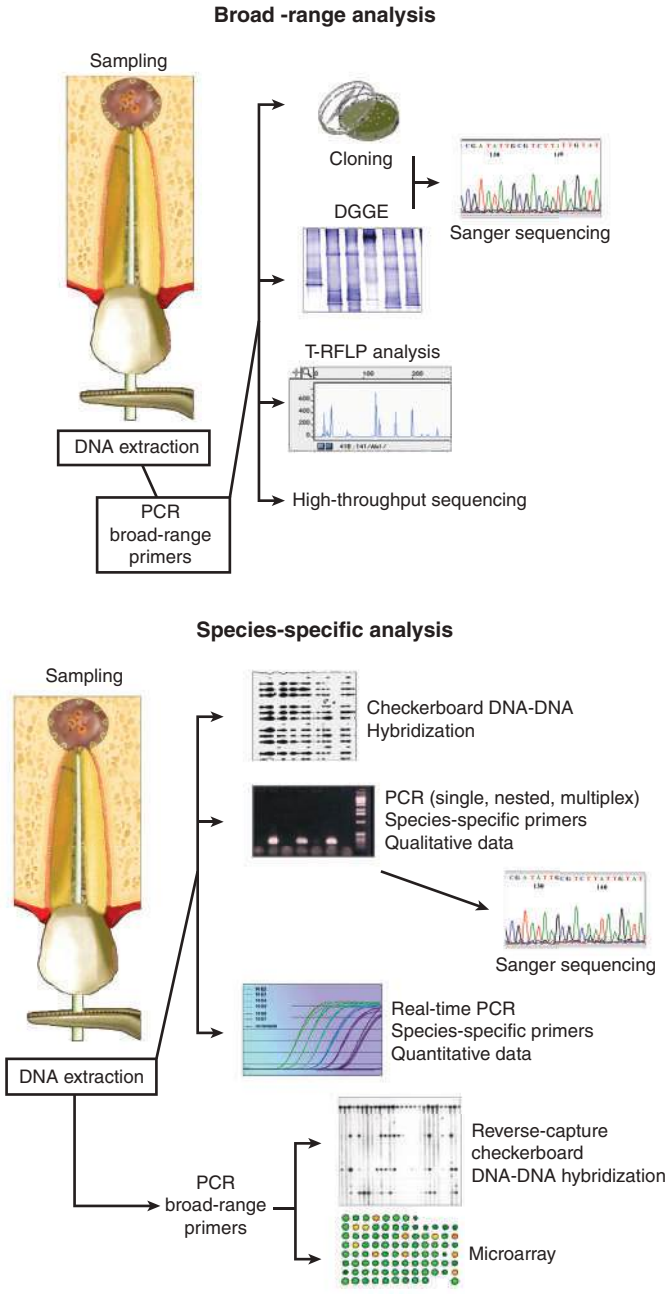


Fig. 15.16 Molecular biology methods used (or with potential to be used) in the study of endodontic infections. The choice for a particular technique will depend on the type of analysis to be performed. Some methods can be used for detection of target microbial species or groups and others for a broader analysis of the microbiota.

bacteria, and newly named species.[21,46,149,231,264,280] The 16S rRNA gene (or 16S rDNA) has been the most widely used gene for bacterial identification, while the 18S rRNA gene of fungi and other eukaryotes has also been used extensively to identify these organisms.

Data from 16S rRNA gene sequences can be used for accurate and rapid identification of known and unknown bacterial species, using techniques that do not require cultivation. The 16S RNA gene of virtually all bacterial species in a given environment, including still uncultivated and uncharacterized bacteria, can be amplified by polymerase chain reaction (PCR) using broad-range (or universal) primers that are complementary to conserved regions of this gene. Sequencing of the variable regions flanked by the broad-range primers will provide information for accurate bacterial identification. Primers or probes that are complementary to variable regions can also be designed to detect specific target species directly in clinical samples.

Many molecular methods for the study of microorganisms exist; the choice of a particular approach depends on the questions being addressed.[240] Broad-range PCR followed by cloning and Sanger sequencing can be used to unravel the breadth of bacterial diversity in a given environment. High-throughput (or "next-generation") sequencing (HTS) approaches permit DNA sequencing from samples to a far deeper coverage and higher throughput in comparison with the traditional Sanger sequencing approach[82,294]; even low-abundant components of the bacterial community can be disclosed by HTS approaches. In addition to HTS, bacterial community structures can be analyzed and compared by fingerprinting techniques such as denaturing gradient gel electrophoresis (DGGE) and terminal restriction fragment length polymorphism (T-RFLP). Fluorescence in situ hybridization (FISH) can measure the abundance of target species and provide information on their spatial distribution in tissues. Among other applications, DNA-DNA hybridization arrays (checkerboard techniques, DNA microarrays), species-specific single PCR, nested PCR, multiplex PCR, and quantitative real-time PCR can be used to survey large numbers of clinical samples for the presence of target species. Quantitative real-time PCR can also be applied to the study of the antibacterial effectiveness of treatment procedures by quantifying all bacteria or certain groups before and after treatment. Variations in PCR technology can also be used to type microbial strains. As with any other method, molecular methods have their own advantages and limitations (Box 15.2).

Box 15.2 Advantages and Limitations of Molecular Biology Methods

Advantages	Limitations
Detect both cultivable and as-yet-uncultivated species or strains	Most assays are qualitative or semiquantitative (exception: real-time PCR)
High specificity and accurate identification of strains with ambiguous or aberrant phenotypic behavior	Most assays only detect one species or a few different species at a time (exceptions: broad-range PCR, checkerboard, microarray)
Detect species directly in clinical samples	Most assays detect only the target species and fail to detect unexpected species (exception: broad-range PCR)
High sensitivity	Some assays can be laborious and costly (e.g., broad-range PCR)
Rapid; most assays take no more than minutes to a few hours to identify a microbial species	Biases in broad-range PCR introduced by homogenization procedures, preferential DNA amplification, and differential DNA extraction
Do not require carefully controlled anaerobic conditions during sampling and transportation	Hybridization assays using whole genome probes detect only cultivable species
Can be used during antimicrobial treatment	Can be very expensive
Anaerobic handling and expertise not required	
Samples can be stored frozen for later analysis	
DNA can be transported easily between laboratories	
Detect dead microorganisms	

PCR, Polymerase chain reaction.

THE FIVE GENERATIONS OF ENDODONTIC MICROBIOLOGY STUDIES

Microbiologic studies for identification of the species participating in endodontic infections can be chronologically divided into five generations on the basis of the different strategic approaches used.[255] These generations are detailed in Table 15.1.

IMPACT OF MOLECULAR METHODS IN ENDODONTIC MICROBIOLOGY

Culture studies (first generation) identified a set of species thought to play an important role in the pathogenesis of apical periodontitis.[11,209,276] Further, not only have findings from culture-based methods been confirmed, but they have also been significantly supplemented with those from culture-independent molecular biology techniques, which constitute the other four generations of endodontic microbiology studies.[236] Molecular methods have confirmed and strengthened the association of many cultivable bacterial species with apical periodontitis and have also revealed new suspected endodontic pathogens.[232] The list of candidate pathogens has expanded to include culture-difficult species or as-yet-uncultivated bacteria that had never been found in endodontic infections by culture. The results from molecular studies impact remarkably on the knowledge of bacterial diversity in endodontic infections. More than 400 different bacterial species have been detected in different types of endodontic infections.[236] Of these, about 45% were exclusively reported by molecular biology studies, compared with 32% detected by culture studies alone.[236] Twenty-three percent of the total bacterial species richness have been detected by application of both culture and molecular studies (Fig. 15.17). As a consequence, it becomes quite evident that the knowledge about the composition of the endodontic microbiota has been refined and redefined by molecular methods.[232]

Types of Endodontic Infections

Endodontic infections can be classified according to the anatomic location as intraradicular or extraradicular infection. *Intraradicular infection* is caused by microorganisms

Table 15.1 Generations of Studies for Microbiologic Identification in Endodontic Infections

Study Generation	Identification Method	Nature	Description and Findings
First	Culture	Open ended (broad range)	Revealed many cultivable species in association with apical periodontitis
Second	Molecular methods (e.g., PCR and its derivatives, original checkerboard assay)	Closed ended (species specific)	Target cultivable bacteria Confirmed and strengthened data from first generation Allowed inclusion of some culture-difficult species in the set of candidate endodontic pathogens
Third	Molecular methods (e.g., PCR-cloning-sequencing, T-RFLP)	Open ended (broad range)	Allowed a more comprehensive investigation of the bacterial diversity in endodontic infections Not only cultivable species but also as-yet-uncultivated and uncharacterized bacteria were identified
Fourth	Molecular methods (e.g., PCR, microarrays, reverse-capture checkerboard)	Closed ended (species specific)	Target cultivable and as-yet-uncultivated bacteria Large-scale clinical studies to investigate prevalence and association of species/phylotypes with endodontic infections
Fifth	Molecular methods (high-throughput sequencing platforms, such as Roche 454 GS-FLX pyrosequencing, Illumina MiSeq and HiSeq)	Open ended (broad range)	Permit a deep-coverage and more comprehensive analysis of the diversity of endodontic infections

PCR, Polymerase chain reaction.

colonizing the root canal system and can be subdivided into three categories according to the time microorganisms entered the root canal system: *primary infection,* caused by microorganisms that initially invade and colonize the necrotic pulp tissue (initial or "virgin" infection); *secondary infection,* caused by microorganisms not present in the primary infection but introduced in the root canal at some time after professional intervention (i.e., *secondary* to intervention); and *persistent infection,* caused by microorganisms that were members of a primary or secondary infection and in some way resisted intracanal antimicrobial procedures and were able to endure periods of nutrient deprivation in treated canals. *Extraradicular infection* in turn is characterized by microbial invasion of the inflamed periradicular tissues and is a sequel to the intraradicular infection. Extraradicular infections may be dependent on or independent of the intraradicular infection.

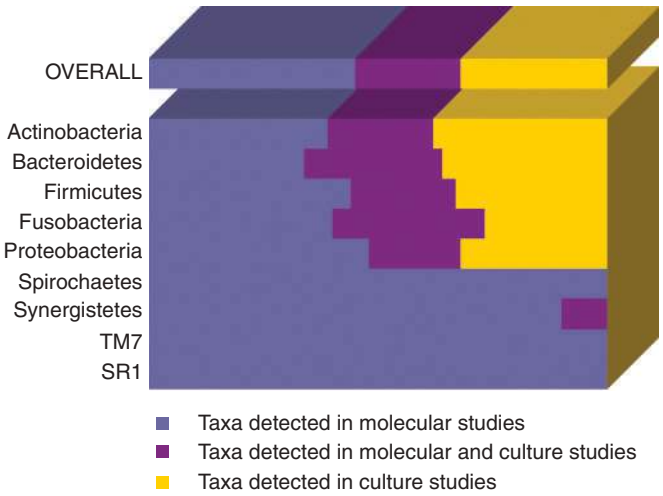


Fig. 15.17 Percentage distribution of bacterial species/phylotypes found in endodontic infections according to the detection method. Data refer to the percentage of species or phylotypes overall or in each of the nine phyla that have endodontic representatives.

Diversity of the Endodontic Microbiota

The terms *microbiota* and *microbiome* have been used interchangeably in many texts, while others have used some confounding definitions. Here, we refer to *microbiome* as a collection of genomes from all the microorganisms in a given site, and *microbiota* as all the microorganisms in a site. Microbiota should replace terms such as *flora* and *microflora,* which perpetuate an outdated classification of microorganisms as plants.[41] *Diversity* refers to the number of different species present (richness) and their relative abundance (evenness) in a given ecosystem.[87]

The oral cavity harbors one of the highest accumulations of microorganisms in the human body. Even though viruses, archaea, fungi, and protozoa can be found as constituents of the oral microbiota, bacteria are by far the most dominant inhabitants of the oral cavity. Culture-independent studies (microscopy and molecular biology technologies) have shown that over 40% to 60% of the oral microbiota still remains to be cultivated and fully characterized.[2,42,262] A high diversity of bacterial species has been revealed in the oral cavity by culturing approaches,[125] but application of molecular biology methods to the analysis of the bacterial diversity has revealed a still broader and more diverse spectrum of extant oral bacteria.[144] More than 1000 bacterial species/phylotypes have been found in studies of the human oral microbiome,[42] but advanced HTS studies have indicated that this number may still have been largely underestimated.[4,71,93] Anaerobic culture and molecular biology techniques have demonstrated that, at a broader taxonomic level, endodontic bacteria fall into nine phyla—namely, Firmicutes, Bacteroidetes, Spirochaetes, Fusobacteria, Actinobacteria, Proteobacteria, Synergistetes, TM7, and SR1

(Fig. 15.18).[126,156,177,195–197,233,236] However, data from studies using HTS technology reveal that several other bacterial phyla may have been overlooked by previous identification techniques, including Tenericutes, Acidobacteria, Deinococcus-Thermus, Chloroflexi, and Cyanobacteria.[63,83,86,95,105,200,213,219,220,224,286,289,309] Members of these less common phyla are usually low-abundant components of the endodontic bacterial communities, only detected because of deep sequencing coverage. Fungi and archaea are types of microorganisms that have been only occasionally found in endodontic infections.

Endodontic infections develop in a previously sterile place that does not contain a normal microbiota. Any species found has the potential to be an endodontic pathogen or at least play a role in the ecology of the endodontic microbial community. Culture and molecular studies of endodontic infections are cross-sectional in nature and consequently reveal only prevalence of species; thus, only association can be inferred. A role in causation of apical periodontitis is usually surmised on the basis of both frequency of detection and potential pathogenicity (in animal models or association with other human diseases), and several species have emerged as candidate endodontic pathogens. The following sections discuss specific aspects of each type of endodontic infection.



Fig. 15.19 Apical periodontitis due to primary intraradicular infection. The dental pulp is necrotic and the lesion size is usually directly proportional to the complexity of the microbiota involved.

Primary Intraradicular Infection

MICROBIAL DIVERSITY

Primary intraradicular infection develops in the necrotic pulp tissue of untreated teeth and is the cause of primary apical periodontitis (Fig. 15.19). Participating bacteria may be involved in earlier stages of pulp invasion (usually via caries) that culminated in inflammation and further necrosis, or they can be latecomers that took advantage of the environmental conditions in the root canal after pulp necrosis.

Primary infections are characterized by a mixed (multispecies) community conspicuously dominated by anaerobic bacteria. The number of bacterial cells may vary from 10^3 to 10^8 per root canal.[19,130,197,249,273,292] Second- to fourth-generation molecular studies have disclosed a mean

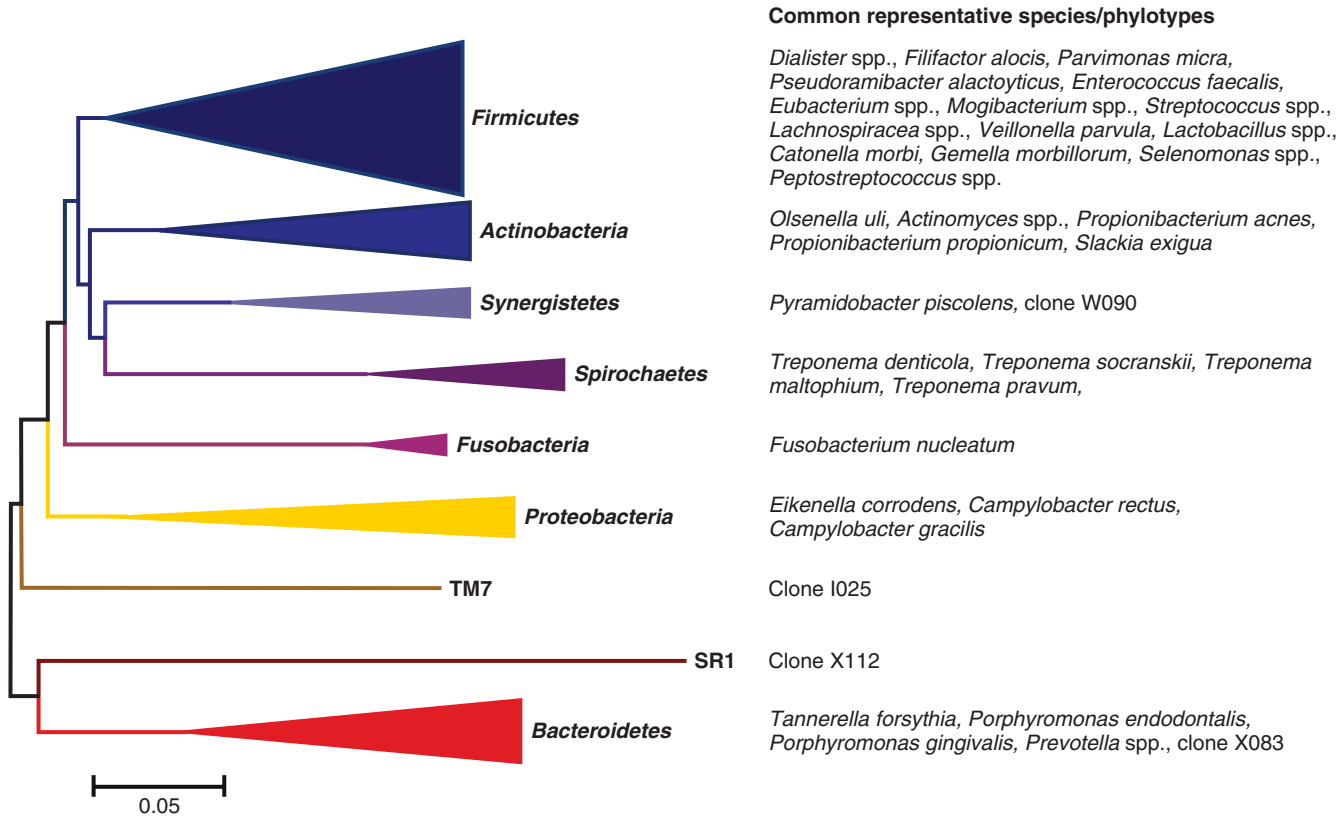


Fig. 15.18 Bacterial phyla with their respective endodontic representatives. *Right:* Example species or phylotypes for each phylum.

of 10 to 20 species/phylotypes per infected canal.[126,156,177,232,251] However, these numbers may be even higher based on fifth-generation HTS studies.[83,105,200,219] Canals of teeth with associated sinus tracts may exhibit a number of species close to the top of this range.[177] The size of apical periodontitis lesion has been shown to be proportional to the number of bacterial species (richness) and cells (density or load) in the root canal.[177,250,273] A molecular study[177] demonstrated that the number of taxa per canal was clearly in direct proportion to the lesion size: small lesions (<5 mm) harbored about 12 taxa, lesions from 5 to less than 10 mm harbored 16 taxa, and lesions over 10 mm harbored about 20 species. Some canals associated with large lesions may harbor even more than 40 taxa.[177] Therefore, the larger the lesion, the higher the bacterial diversity and density in the canal.

The most prevalent named bacterial species detected in primary infections, including acute abscess cases, belong to diverse genera of gram-negative (i.e., *Fusobacterium, Dialister, Porphyromonas, Prevotella, Tannerella, Treponema, Pyramidobacter*, *Campylobacter,* and *Veillonella*) and gram-positive (*Parvimonas, Filifactor, Pseudoramibacter, Streptococcus, Propionibacterium, Olsenella,* and *Actinomyces*) bacteria.[14,55,56,66,74,96,126,156,177,179,183,195,196,232,250,252,253,273,275,293] Bacterial prevalence in primary infections may vary from study to study as a function of several factors, such as sensitivity and specificity of the detection and identification methods, sampling technique, geographic location, and accuracy or divergence in clinical diagnosis and disease classification. Even so, the species most frequently detected are expected to be the same in most well-conducted studies. Figs. 15.20–15.22 display the most frequently detected species associated with asymptomatic apical periodontitis, symptomatic apical periodontitis, and acute apical abscesses, as revealed by studies from the authors' group using a highly sensitive molecular biology technique (nested PCR).

Approximately 40% to 66% of the endodontic microbiota in primary infections is composed of species still uncultivated.[126,156,196] As for their abundance in these infections, as-yet-uncultivated phylotypes corresponded to about 40% of the clones sequenced.[196] Molecular studies investigating the breadth of bacterial diversity in infected root canals have disclosed the occurrence of uncultivated phylotypes belonging

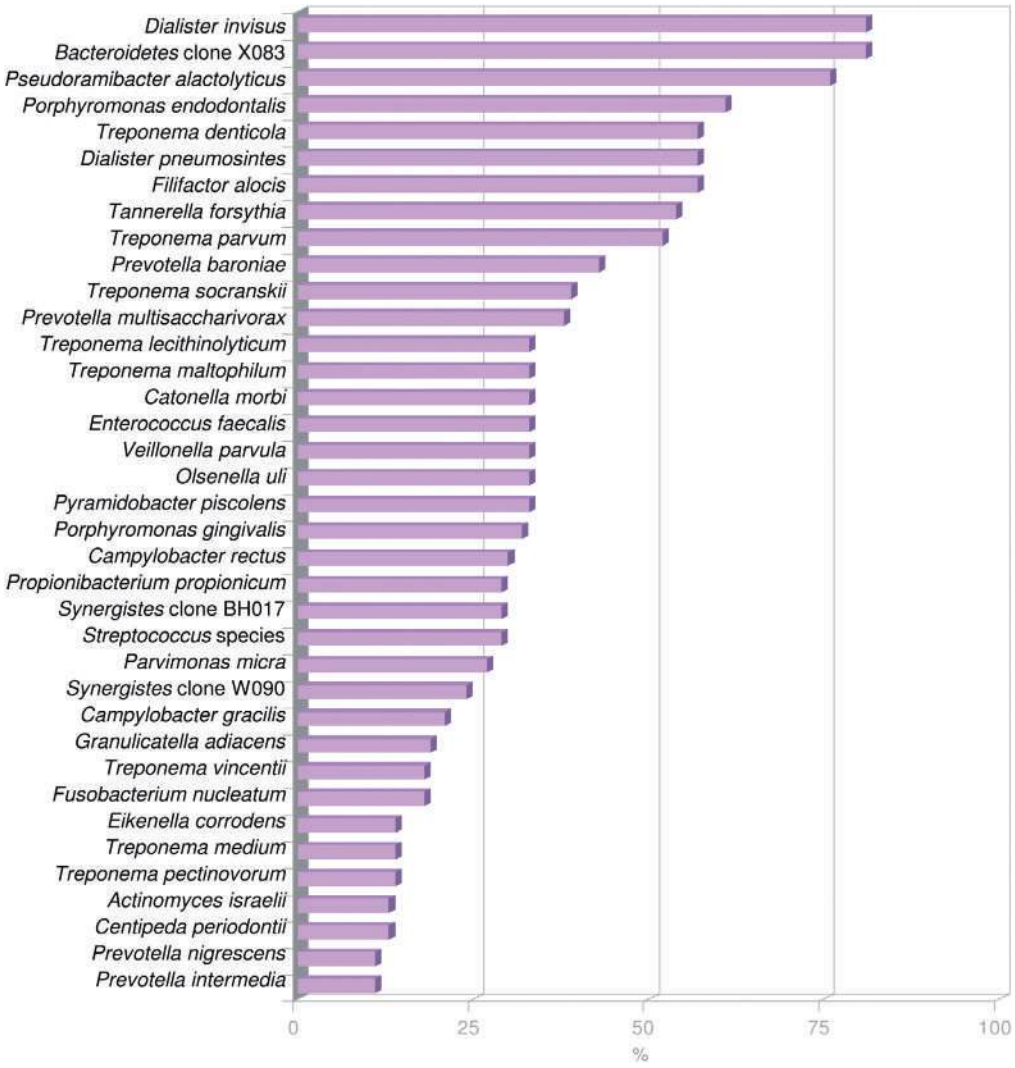


Fig. 15.20 Prevalence of bacteria detected in primary infections of teeth with asymptomatic (chronic) apical periodontitis. Data from the authors' findings using a taxon-specific nested-polymerase chain reaction protocol.

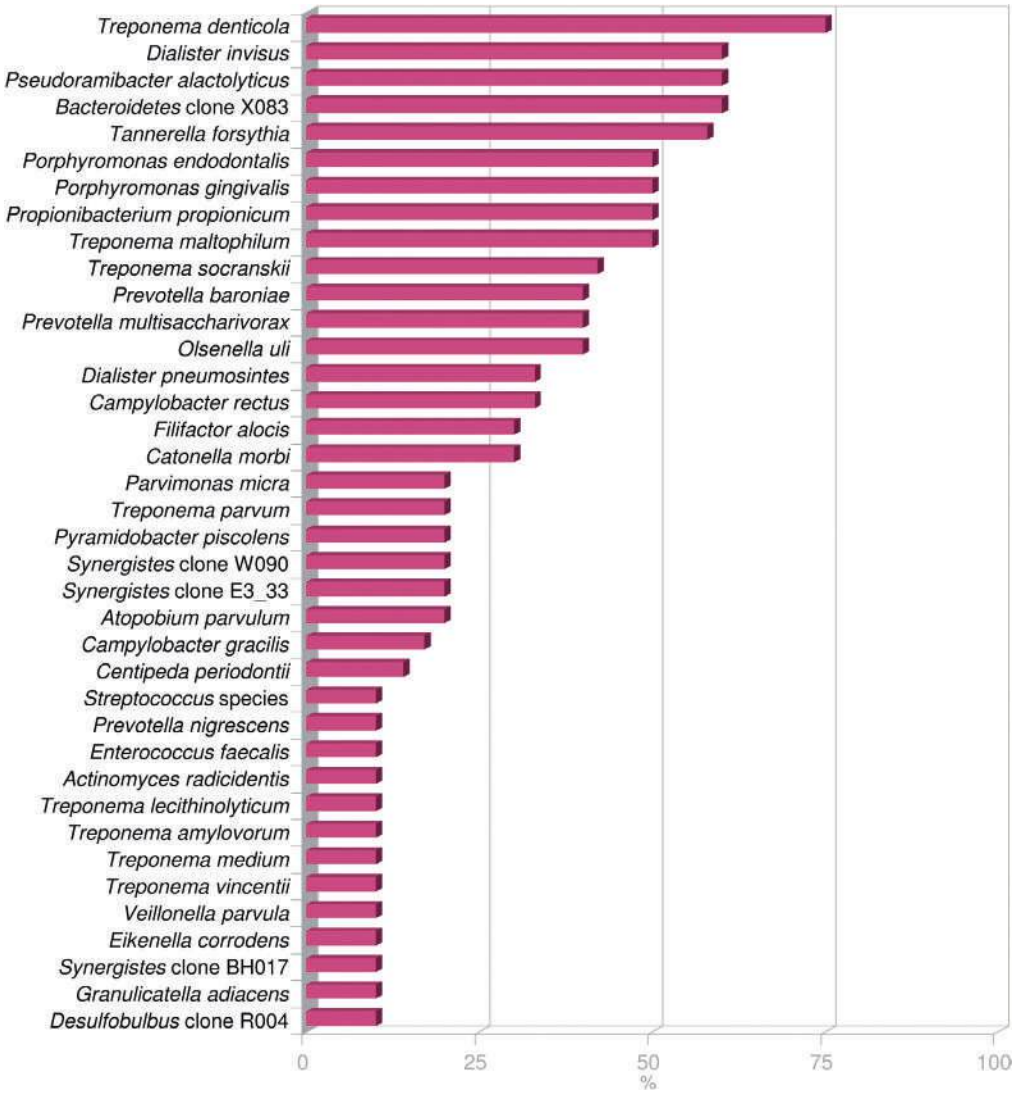


Fig. 15.21 Prevalence of bacteria detected in primary infections of teeth with symptomatic (acute) apical periodontitis. Data from the authors' findings using a taxon-specific nested-polymerase chain reaction protocol.

to several genera, including *Dialister, Prevotella, Solobacterium, Olsenella, Fusobacterium, Treponema, Eubacterium, Megasphaera, Veillonella,* and *Selenomonas,* as well as phylotypes related to the family *Lachnospiraceae* or the TM7 and Synergistetes phyla.[126,156,175,176,188,195,196,199,233,243] Some uncultivated phylotypes can even be among the most prevalent bacteria in primary intraradicular infections, and others may be associated with pain.[196] *Bacteroidaceae* sp. HOT-272 (synonym, Bacteroidetes oral clone X083) is one of the most prevalent phylotypes found in endodontic infections.[177–179,241] Detection of as-yet-uncultivated phylotypes in samples from endodontic infections suggests that they can be previously unrecognized bacteria that play a role in the pathogenesis of different forms of apical periodontitis. The fact that they have not yet been cultivated and phenotypically characterized does not mean that they are not important.[238]

SYMPTOMATIC INFECTIONS

Symptomatic apical periodontitis and acute apical abscesses are typical examples of endodontic infections causing severe symptoms. In these cases, the infection is located in the canal, but it has also reached the periradicular tissues and, in abscessed cases, can spread to other anatomic spaces. Acute apical abscesses are caused by bacteria that egress from the infected root canal and invade the periradicular tissues to establish an extraradicular infection and evoke purulent inflammation. Clinically, the disease leads to pain or swelling and has the potential to diffuse to sinuses and fascial spaces of the head and neck to form a cellulitis or other complications (Fig. 15.23). The microbiota involved in endodontic abscesses is mixed and dominated by anaerobic bacteria (see Fig. 15.22).[37,96,100,169,196,237,239,251] Direct comparisons using molecular technology revealed an average of 12 to 18 taxa per abscess case, compared with 7 to 12 taxa present in root canals of teeth with asymptomatic lesions.[196,251] Uncultivated phylotypes constitute approximately 40% of the taxa found in abscesses (richness) and collectively represent more than 30% of the 16S rRNA gene sequences retrieved in clone libraries (abundance).[196] The total number of bacterial cells in acute apical abscesses has been shown to vary from 10^4 to 10^9.[96,103,305]

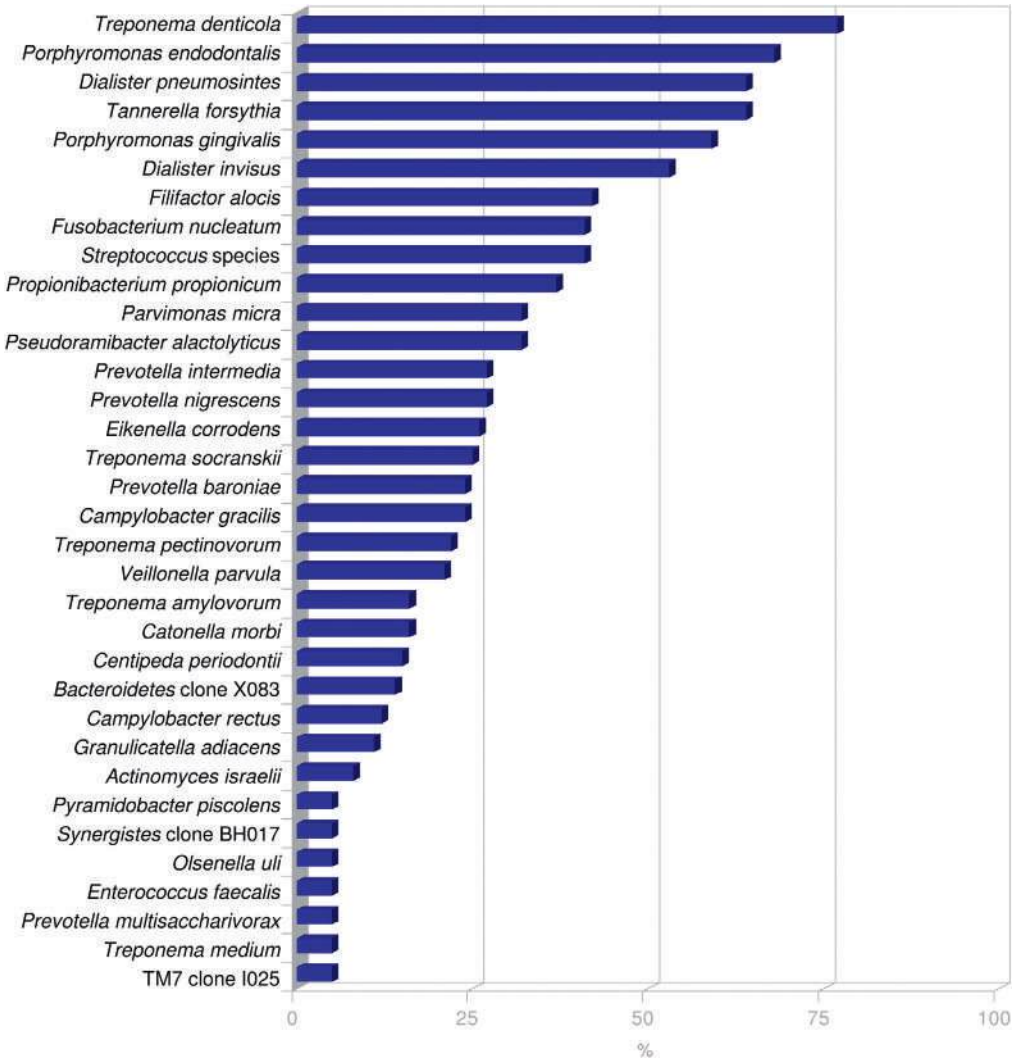


Fig. 15.22 Prevalence of bacteria detected in primary infections of teeth with acute apical abscesses. Data from the authors' findings using a taxon-specific nested-polymerase chain reaction protocol.



Fig. 15.23 Acute apical abscess. The infection has spread to other anatomic spaces to form cellulitis. (Courtesy Dr. Henrique Martins.)

Whereas microbial causation of apical periodontitis is well established, there is no strong evidence disclosing specific involvement of a single species with any particular sign or symptom of apical periodontitis. Some gram-negative anaerobic bacteria have been suggested to be involved with symptomatic lesions,[66,70,185,196,273,288,308] but the same species may also be present in somewhat similar frequencies in asymptomatic cases.[14,56,74,90,183,252,253] Thus, factors other than the mere presence of a given putative pathogenic species may play a role in the etiology of symptomatic endodontic infections.[214,221,239] These factors include: differences in virulence ability among strains of the same species, bacterial interactions resulting in synergistic effects among species in mixed infections, overall number of bacterial cells (overall infectious load), counts and relative abundance of specific pathogenic species (specific infectious load), environmental cues regulating expression of virulence factors, host resistance, and concomitant herpesvirus infection. Association of some or all of these factors is likely to determine the occurrence and intensity of symptoms.[214,221]

Molecular studies using DGGE, T-RFLP, or HTS analysis have revealed that the structure of the endodontic bacterial communities in symptomatic teeth, including abscess cases, is significantly different from that of asymptomatic teeth.[196,200,251] Differences are represented by different dominant species in the communities and larger numbers of species in symptomatic cases. Differences in the type and load of dominant species and the resulting bacterial interactions may affect virulence of the whole bacterial community. The counts of some specific pathogenic species have been shown to be important for the community pathogenicity in abscesses. Molecular studies have shown that the levels of *Porphyromonas endodontalis, Prevotella baroniae, Tannerella forsythia, Treponema denticola,* and *Streptococcus* species were significantly higher in samples from abscesses than in asymptomatic teeth.[183,202] In addition, different species combinations may result in different outcomes because of the network of interactions.[186] The possibility exists that bacterial interactions result in communities that are more or less aggressive and consequently can cause host responses of corresponding intensity. This is consistent with the concept that the entire bacterial community is the unit of pathogenicity.

GEOGRAPHIC INFLUENCE

Findings from laboratories in different countries are often different regarding the prevalence of the species involved in endodontic infections. Although these differences may be attributed to technical issues, a geographic influence in the composition of the root canal microbiota has been suspected. Studies using molecular biology techniques directly compared the prevalence of candidate endodontic pathogens in endodontic samples from patients residing in distinct geographic locations and observed significant differences.[13,171,226] Analysis of the bacterial community profiles of the microbiota associated with endodontic infections from different countries also revealed a geography-related pattern, with several species being exclusive for each location and others shared by the locations showing great differences in prevalence.[113,244] The factors that can lead to differences in the composition of the endodontic microbiota and the impact of these differences on therapy, particularly in abscess cases requiring systemic antibiotic therapy, remain elusive.

MICROBIAL ECOLOGY AND THE ROOT CANAL ECOSYSTEM

A root canal with necrotic pulp provides a space for bacterial colonization and affords bacteria a moist, warm, nutritious, and anaerobic environment, which is by and large protected from the host defenses because of lack of an active blood circulation in the necrotic pulp tissue. Also, the root canal walls are nonshedding surfaces conducive to persistent colonization and formation of complex communities. The necrotic root canal might be considered a fertile environment for bacterial growth, with colonization not being a difficult task for most oral bacterial species. Although a large number of bacterial species (about 100 to 200) can be found in the oral cavity of a particular individual,[144] only a limited assortment of these species (about 10 to 20) is consistently selected out for growth and survival within a root canal containing necrotic pulp tissue from the same individual. This indicates that ecologic determinants operate in the necrotic canal and dictate which species will succeed in colonizing this previously sterile environment. The major ecologic factors that determine the composition of the root canal microbiota include oxygen tension, type and amount of available nutrients, and bacterial interactions. Other factors such as temperature, pH, and receptors for adhesins may also be involved.

The root canal infection is a dynamic process, and different bacterial species apparently dominate at different stages.[281] Shifts in the composition of the microbiota are largely due to changes in environmental conditions, particularly in regard to oxygen tension and nutrient availability. In the initial phases of the pulpal infectious process, facultative bacteria predominate.[51] After a few days or weeks, oxygen is depleted within the root canal as a result of pulp necrosis and consumption by facultative bacteria. Further oxygen supply is interrupted with loss of blood circulation in the necrotic pulp. An anaerobic milieu develops and is highly conducive to the survival and growth of obligate anaerobic bacteria. With the passage of time, anaerobic conditions become even more pronounced, particularly in the apical third of the root canal; as a consequence, anaerobes will dominate the microbiota, outnumbering facultative bacteria (Fig. 15.24).

The main sources of nutrients for bacteria colonizing the root canal system include (1) the necrotic pulp tissue, (2) proteins and glycoproteins from tissue fluids and exudate that seep into the root canal system via apical and lateral foramina, (3) components of saliva that may coronally penetrate into the root canal, and (4) products of the metabolism of other bacteria. Because the largest amount of nutrients is available in the main canal, which is the most voluminous part of the root canal system, the highest numbers of bacteria (particularly fastidious anaerobic species) are expected to be located in this region. Bacterial species that can best utilize and compete for nutrients in the root canal system will succeed in colonization.

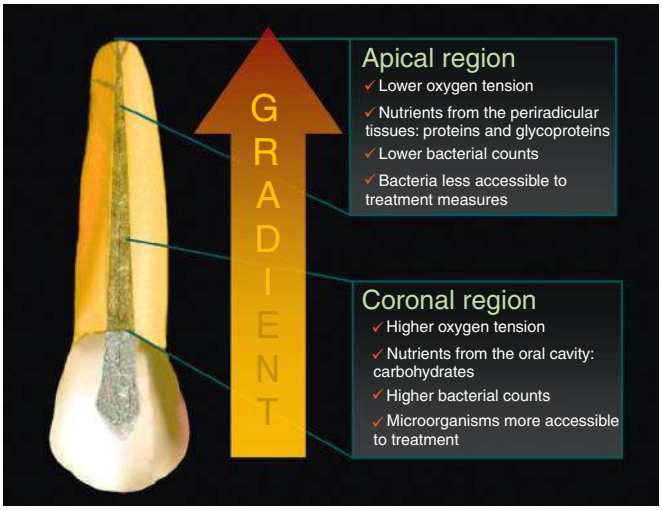


Fig. 15.24 Ecologic conditions in different areas of the root canal. A gradient of oxygen tension and nutrients (type and availability) is formed. Consequently, the microbiota residing in different parts can also differ in diversity, density, and accessibility to treatment procedures.

In addition to being influenced by variations in oxygen levels, shifts in the composition of the microbiota colonizing the root canal system can also be dependent upon the dynamics of nutrient utilization. Saccharolytic species dominate the very early stages of the infectious process but are soon outnumbered by asaccharolytic species, which will dominate later stages. Even though the necrotic pulp tissue can be regarded as a finite source of nutrients to bacteria (given the small volume of tissue that is progressively degraded), induction of periradicular inflammation guarantees a sustainable source of nutrients, particularly in the form of proteins and glycoproteins present in the exudate that seep into the canal. At this stage of the infectious process, bacteria that have a proteolytic capacity or establish a cooperative interaction with those that can utilize this substrate in the metabolism, start to dominate.[278] Therefore, as the infectious process reaches the stage of induction of periradicular inflammation, proteins become the principal nutrient source, particularly in the apical part of the canal, favoring the establishment of anaerobic species that utilize peptides or amino acids in their metabolism (see Fig. 15.24).

Because primary endodontic infections are usually characterized by mixed communities, different bacterial species are in close proximity with one another, and interactions become inevitable. Thus, establishment of certain species in the root canal is also influenced by interactions with other species. In this regard, early colonizers play an important role in dictating which species will live along with them in the community. Bacterial interactions can be positive or negative. Positive interactions enhance the survival capacity of the interacting bacteria and enable different species to coexist in habitats where neither could exist alone. For instance, interbacterial nutritional interactions are important ecologic determinants that result in higher metabolic efficiency of the whole community. Nutritional interactions are mainly represented by food webs, including utilization of metabolic end products from one species by another and bacterial cooperation for the breakdown of complex host-derived substrates. Moreover, one species can provide growth conditions favorable to another—for example, by reducing oxygen tension in the environment and favoring the establishment of anaerobes, or by releasing proteinases that can provide protection from host defenses. Negative interactions in turn act as feedback mechanisms that limit population densities. Examples include competition (for nutrients and space) and amensalism (or antagonism, when one species produces a substance that inhibits another species, e.g., bacteriocins).

Many species adhere directly to host surfaces, whereas other species adhere to bacteria already attached to the surface. The latter is called *coaggregation* and is a highly specific phenomenon with regard to the partners involved.[98] A given pair of species can attach to each other by means of specific receptor-adhesin interactions, which are usually lectin-like interactions (attachment of a specific protein on the surface of one species to a specific carbohydrate on the surface of the other). Coaggregation can favor colonization of host surfaces and also facilitate metabolic interactions between the partners. Coaggregation has been demonstrated for several pairs of bacterial taxa found in endodontic infections.[97]

OTHER MICROORGANISMS IN ENDODONTIC INFECTIONS

Fungi

Fungi are eukaryotic microorganisms that may colonize the oral cavity, especially *Candida* species, but they have been only occasionally detected in primary intraradicular infections.[49,101,122,248] Fungi are more frequently detected in canals of teeth with posttreatment disease (discussed later).

Archaea

Archaea consist of a highly diverse group of prokaryotes distinct from bacteria. Members of this domain have been traditionally recognized as extremophiles, but some of these microorganisms have also been found to flourish in nonextreme environments, including the human body.[48] No member of the *Archaea* domain has been described as a human pathogen. Several studies have failed to detect archaea in samples from primary endodontic infections.[180,181,242] However, other studies have reported on the detection of these organisms, including a *Methanobrevibacter oralis*–like phylotype, in some cases.[136,137,291,293] Because archaea have not been consistently detected in infected root canals, their role, if any, in the pathogenesis of apical periodontitis is questionable.

Viruses

Viruses are inanimate particles composed of a nucleic acid molecule (DNA or RNA) and a protein coat. On their own, they have no metabolism. In order to replicate the viral genome, they need to infect living cells and use the cell's machinery; as a consequence, they cannot prosper in the necrotic root canal. Viruses have been detected in root canals only in teeth with vital pulps. A study found the human immunodeficiency virus (HIV) in vital pulps of HIV-seropositive individuals.[62] Another study reported the occurrence of some herpesviruses in normal and inflamed vital pulps.[104]

Apical periodontitis lesions, in turn, are characterized by dense accumulations of viable immune and connective tissue cells. Sabeti and Slots first reported on the occurrence of herpesviruses in these lesions.[190–193] It was hypothesized that herpesviruses might be implicated in the pathogenesis of apical periodontitis as a direct result of virus infection and replication or as a result of virally induced impairment of local host defenses, which would favor the overgrowth of pathogenic bacteria in the very apical part of the root canal.[261] Bacterial challenge emanating from the canals might cause an influx of virus-infected cells into the periradicular tissues. Reactivation of herpesviruses by tissue injury induced by bacteria would evoke impairment of host immune response in the periradicular microenvironment, changing the potential of local defense cells to mount an adequate response against infectious agents. In addition, herpesviruses may directly stimulate inflammatory cells to release proinflammatory cytokines.[120,301]

Herpesviruses have been found in samples from symptomatic apical periodontitis lesions,[191,193] abscesses,[27,54] large lesions,[192,193] and lesions from HIV-positive patients.[194] Most studies reported on the occurrence of herpesviruses in samples of apical periodontitis, but this cannot be necessarily interpreted as active infection. Herpesvirus-infected inflammatory cells may be attracted to the periradicular

tissues in response to bacterial infection. Once these cells accumulated in the area, herpesviruses can be easily detected. While more consistent information is not available, the role of herpesviruses in the pathogenesis of apical periodontitis, if any, cannot be ascertained.

Persistent/Secondary Endodontic Infections

As defined earlier in this chapter, persistent intraradicular infections are caused by microorganisms that resisted intracanal antimicrobial procedures and survived in the treated canal. Involved microorganisms are usually remnants of a primary infection. The secondary infection, in turn, is caused by microorganisms that at some time entered the root canal system secondary to clinical intervention. The moment can be during treatment, between appointments, or even after root canal filling. In any circumstance, if penetrating microorganisms manage to adapt themselves to the new environment, surviving and flourishing, a secondary infection is established. Species involved can be oral species or not, depending on the source of secondary infection.

Persistent and secondary infections are for the most part clinically indistinguishable. Exceptions include infectious complications (such as an apical abscess) arising after the treatment of noninfected vital pulps or cases in which apical periodontitis was absent at the time of treatment but present on the follow-up radiograph. Both situations are typical examples of secondary infections. Both persistent and secondary infections can be responsible for several clinical problems, including persistent exudation ("wet" canal), persistent symptoms, interappointment flare-ups, and failure of the endodontic treatment, characterized by a posttreatment apical periodontitis lesion.

PERSISTENT/SECONDARY INFECTIONS AND TREATMENT FAILURE

Intraradicular infections, either persistent or secondary, can be regarded as the major causes of posttreatment apical periodontitis (Fig. 15.25). Other causes include extraradicular infections (discussed later in this chapter) and nonmicrobial factors, for which there is only weak evidence coming from case reports and speculation.[129] The major role of persistent/secondary infections in treatment failure is supported by two strong evidence-based arguments. First, it has been demonstrated that there is an increased risk of adverse treatment outcome when bacteria are present in the canal at the time of obturation.[52,297] Second, virtually all root canal–treated teeth evincing persistent apical periodontitis lesions have been demonstrated to harbor an intraradicular infection.[106,108,150,167,173,198,230,277] Based on these arguments, studies investigating bacteria remaining in the root canals at the obturation stage disclose species that have the potential to influence the treatment outcome *(predicting the outcome)*. On the other hand, studies dealing with the microbiota of root canal–treated teeth with apical periodontitis show the association of species with treatment failure, because the microorganisms detected are likely to be the cause of posttreatment disease *(outcome established)*.



Fig. 15.25 Posttreatment apical periodontitis lesions in root canal–treated teeth. In inadequately treated canals, the microbiota is similar to primary infections. In cases apparently well treated, fewer species are found. Regardless of treatment quality, persistent or secondary intraradicular infections are the main causative agents of endodontic treatment failure.

BACTERIA AT THE ROOT CANAL OBTURATION STAGE

Diligent antimicrobial treatment may still fail to completely eliminate bacteria from the infected root canal system.[235] This is mostly because persisting bacteria are located in areas inaccessible to treatment procedures,[247] although resistance to some substances have been suggested as cause of bacterial persistence.[57,300] Whatever the cause of persistence, bacterial diversity and density in infected canals are substantially reduced after treatment. Root canal samples positive for bacterial growth after chemomechanical procedures, followed or not by intracanal medication, have been shown to harbor one to five bacterial species per case, and the number of persistent bacterial cells usually varies from 10^2 to 10^5 per sample.[9,22,130,174,197,228,229,292]

No single species has been significantly found to persist after treatment procedures. Gram-negative bacteria, which are common members of primary infections, are usually eliminated. Exceptions include some anaerobic rods, such as *F. nucleatum* and *Prevotella* species, which are among the species found in postinstrumentation or postmedication samples.[22,67,147,197,249,258,309] Most studies on this subject have clearly revealed that when bacteria resist treatment procedures, gram-positive bacteria are more frequently present. Gram-positive facultatives or anaerobes often detected in these samples include streptococci, *P. micra, Propionibacterium* species, *P. alactolyticus, Actinomyces* species, lactobacilli, *E. faecalis,* and *Olsenella uli*.[22,24–26,29,67,145,147,179–181,197,225,228,229,258,279] This supports the notion that gram-positive bacteria can be more resistant to antimicrobial treatment measures and have the ability to adapt to the harsh environmental conditions in instrumented and medicated root canals. As-yet-uncultivated bacteria have also been detected in posttreatment samples,[138,197] indicating that they may also resist antimicrobial treatment.

Bacteria persisting in the root canal after chemomechanical procedures or intracanal medication will not always maintain an infectious process. This statement is supported by evidence that some apical periodontitis lesions healed even after bacteria were isolated from the canal at the obturation stage.[52,258] There are some possible explanations[235]:

- Residual bacteria may die after obturation because of toxic effects of the filling material, access denied to nutrients, or disruption of bacterial ecology.
- Residual bacteria may be present in quantities and virulence subcritical to sustaining periradicular inflammation.
- Residual bacteria remain in locations of the root canal system where access to periradicular tissues is denied.

Bacteria that resist intracanal procedures and are present in the canal at the time of obturation can influence the outcome of the endodontic treatment provided they do the following[235]:

- Have the ability to withstand periods of starvation, scavenging for low traces of nutrients or assuming a dormant state or a state of low metabolic activity (viable but noncultivable -VBNC), to prosper again when the nutrient source is reestablished.
- Resist treatment-induced disturbances in the ecology of the bacterial community, including disruption of quorum-sensing systems, food webs, genetic exchanges, and disorganization of protective biofilm structures.
- Reach a critical population density (load) necessary to inflict damage to the host.
- Have unrestrained access to the periradicular tissues through apical/lateral foramina or iatrogenic root perforations.
- Possess virulence attributes that are expressed in the modified environment and reach concentrations adequate to directly or indirectly induce damage to the periradicular tissues.
- In this context, the host resistance to infection is also an important and decisive counteracting factor.

MICROBIOTA IN ROOT CANAL–TREATED TEETH

The microbiota in root canal–treated teeth with apical periodontitis exhibits decreased diversity in comparison to primary infections. In addition, canals apparently well treated harbor less species than canals with inadequate treatment; the latter can exhibit similar species richness to untreated canals.[150,184,198,230,277] The bacterial density in treated teeth with posttreatment disease varies from 10^3 to 10^7 per canal.[19,145,182,207,310] Likewise, inadequately treated canals are expected to have a higher bacterial density than well-treated canals.

Several culture and molecular biology studies have revealed that *E. faecalis* is the most frequent species in root canal–treated teeth, with prevalence values in some studies reaching more than 90% of cases (Fig. 15.26).[121,150,173,182,187,207,230,277,312] Rôças et al.[187] reported that root canal–treated teeth are about nine times more likely to harbor *E. faecalis* than cases of primary infections. This suggests that other members of a mixed bacterial consortium commonly present in primary infections can inhibit this species, and that the bleak environmental conditions within obturated root canals do not prevent its survival. The fact that *E. faecalis* has been commonly recovered from teeth treated at multiple visits or left open for drainage[257] suggests that this species is a secondary invader capable of colonizing the canal and resisting treatment. In other words, *E. faecalis* may cause secondary infections that later become persistent. *E. faecalis is* considered a transient species in the oral cavity; its source may be food.[311]

For a given microorganism to survive in root canal–treated teeth, it has to resist intracanal disinfection procedures and adapt to the harsh environmental conditions caused by treatment. The ability of *E. faecalis* to penetrate dentinal tubules, sometimes to a deep extent (Fig. 15.27),[73,223] can enable it to escape the action of endodontic instruments and irrigants used during chemomechanical preparation.[73,222] Moreover, its ability to form biofilms in root canals can be important for its resistance to and persistence after intracanal antimicrobial procedures.[43] *Enterococcus faecalis* is also resistant to calcium hydroxide.[73] Finally, environmental cues can regulate gene expression in *E. faecalis,* affording this bacterium the ability to adapt to varying (and adverse) conditions.[89] Indeed, *E. faecalis* can enter a VBNC state,[109] which is a survival mechanism adopted by several bacteria when exposed to unfavorable environmental conditions.[110] In the VBNC state, bacteria lose the ability to grow in culture media but maintain viability and pathogenicity and sometimes can resume division when optimal environmental conditions are restored. *E. faecalis* has the capacity to recover from a prolonged starvation state in obturated canals,[208] suggesting that viable cells of this species, entombed at the time of canal obturation, may provide a long-term nidus for subsequent infection.

Taken together, all these properties help explain the significantly high prevalence of *E. faecalis* in root canal–treated teeth. Although association of this species with posttreatment disease is suggested by cross-sectional studies and supported by the species attributes that allow it to survive under unfavorable environmental conditions, causation is unproved. In fact, the following findings from studies carried out in independent laboratories have questioned the status of *E. faecalis* as the main causative agent of endodontic failures:

- Despite being easily cultivated, *E. faecalis* is not detected in all studies evaluating the microbiota of root canal–treated teeth with posttreatment disease.[28,188]
- In many studies that detected *E. faecalis*, it was rarely one of the most dominant species in the microbiota of re-treatment cases.[81,172,184,198,220]
- *E. faecalis* is found in similar prevalence in root canal–treated teeth with or without apical periodontitis.[92,312]

Other bacteria found in endodontically treated teeth with apical periodontitis include *Streptococcus* species and some fastidious anaerobic species, such as *P. alactolyticus, Propionibacterium* species, *Filifactor alocis, Dialister* species, *Tannerella forsythia, P. micra,* and *Prevotella* species (Fig. 15.28).[7,68,81,121,150,172,182,184,198,230,233,277,309,310] Uncultivated phylotypes correspond to 55% of the taxa detected in treated canals, and collectively they can also be in high proportions, corresponding to about half of the 16S rRNA gene sequences retrieved in clone libraries.[198] Detection of uncultivated phylotypes helps explain why culture studies fail to detect bacteria in some treated root canals.

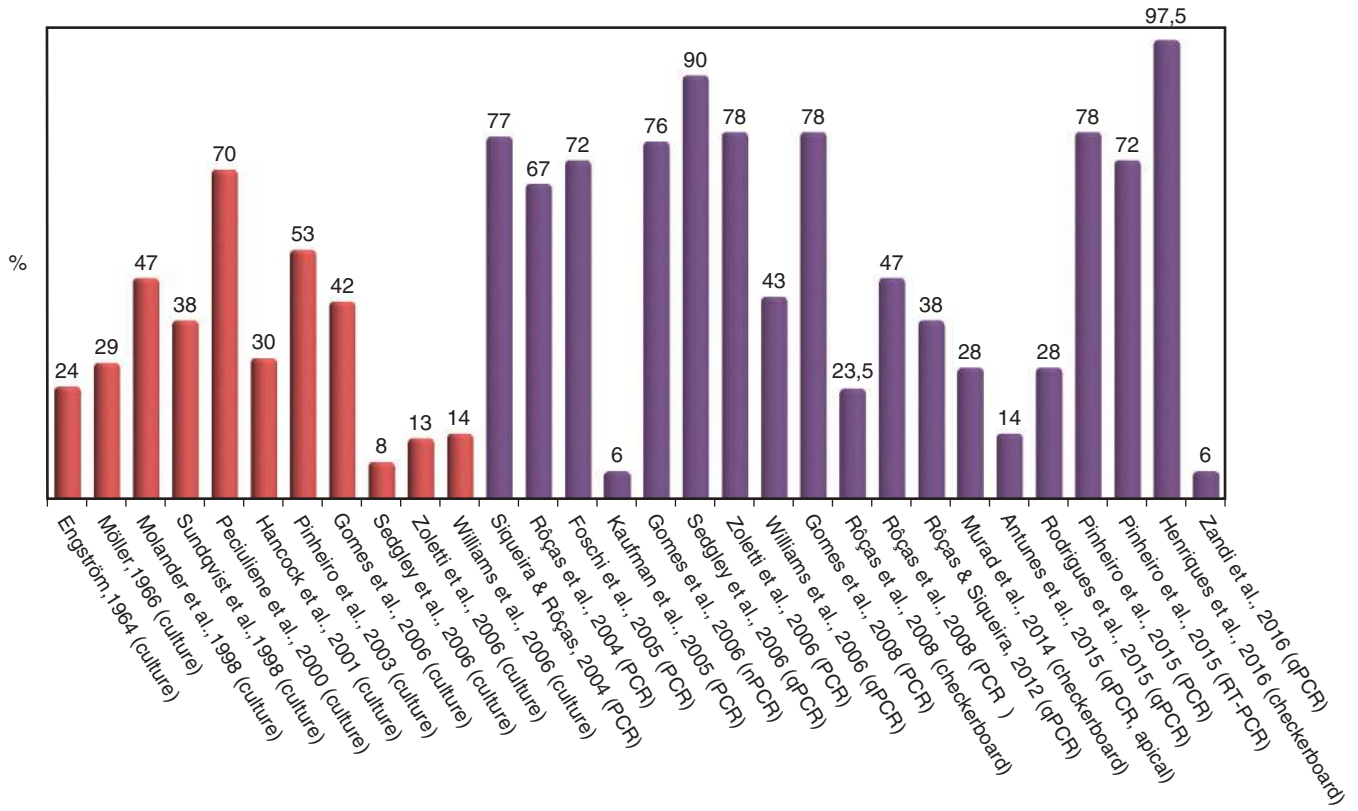


Fig. 15.26 Prevalence of *Enterococcus faecalis* in samples from root canal–treated teeth with apical periodontitis. Data from culture *(red bars)* and molecular *(purple bars)* studies follow.

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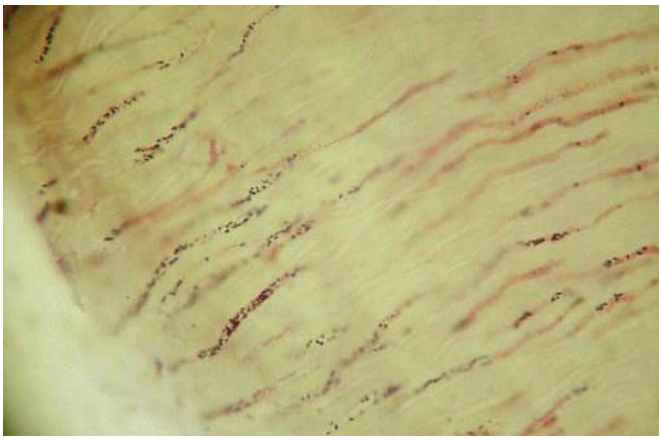


Fig. 15.27 Dentinal tubule infection by *Enterococcus faecalis* in a dog's tooth after experimental infection. Notice that cells invaded the entire extent of some tubules up to the cementum (Brown and Brenn staining, ×1000).

The bacterial community profiles in treated cases vary from individual to individual, indicating that distinct bacterial combinations can play a role in treatment failure (Fig. 15.29).[184,198,220,309] Molecular findings indicate that the microbiota of root canal–treated teeth with apical periodontitis is more complex than previously demonstrated by culture studies.

Fungi are only occasionally found in primary infections, but *Candida* species have been detected in root canal–treated teeth in up to 18% of the cases.[28,49,121,122,145,150,230,277] Fungi gain access to root canals via contamination during endodontic therapy (secondary infection) or they overgrow after inefficient intracanal antimicrobial procedures that cause an imbalance in the primary endodontic microbiota.[254] *Candida albicans* is by far the most commonly detected fungal species in root canal–treated teeth. This species has several properties that can be involved in persistence following treatment, including its ability to colonize and invade dentin[211,212,246] and resistance to calcium hydroxide.[298,300]

Extraradicular Infections

Apical periodontitis lesions are formed in response to intraradicular infection and by and large constitute an effective barrier against spread of the infection to the alveolar bone and other body sites. In most situations, apical periodontitis inflammatory lesions succeed in preventing bacteria from invading the periradicular tissues. Nevertheless, in some specific circumstances, bacteria can overcome this defense barrier and establish an extraradicular infection. The most common form of extraradicular infection is the acute apical abscess, characterized by purulent inflammation in the periradicular tissues in response to a massive egress of virulent bacteria from the root canal. Extraradicular infections in the form of biofilm covering the outer root surface, along with complex bacterial arrangements in the apical canal system, are commonly observed in cases of chronic abscesses (with sinus tract) and "wet canals"(persistent exudation).[157,159] Extraradicular biofilm, adhered to the external root surface around the apical foramina, and apical actinomycosis, characterized by cohesive actinomycotic colonies in the lesion body (Fig. 15.30),[78,132,161,168,283] have been regarded as possible extraradicular causes of post-treatment apical periodontitis.

An extraradicular infection can develop as follows[216]:

- It can be a result of direct advance of some bacterial species that overcome host defenses concentrated near or beyond the apical foramen, an extension of the intraradicular infectious process, or bacterial penetration into the lumen of a cyst that is in direct communication with the apical foramen ("bay" cyst). The borderline between the infecting endodontic microbiota and the host defenses is often located intraradicularly, short of or at the apical foramen. In some cases, however, bacteria may reach the periradicular tissues, and the borderline is then situated extraradicularly, beyond the boundaries of the apical foramen (Fig. 15.31). In the latter situation, the whole infectious process is a continuum composed of an intraradicular and an extraradicular segment, in which the former usually fosters the latter (see Fig. 15.31). It has been suggested that in some cases, the extraradicular segment may become independent of the intraradicular component of the infectious process.[168]
- It can result from bacterial persistence in the apical periodontitis lesion after remission of an acute apical abscess. The acute apical abscess is for the most part clearly dependent on the intraradicular infection; once the intraradicular infection is properly controlled by root canal treatment or tooth extraction and drainage of pus is achieved, the extraradicular infection is handled by the host defenses and usually subsides. Nonetheless, it should be appreciated that in some rare cases, bacteria that have participated in acute apical abscesses may persist in the periradicular tissues following resolution of the acute response and establish a persistent extraradicular infection associated with

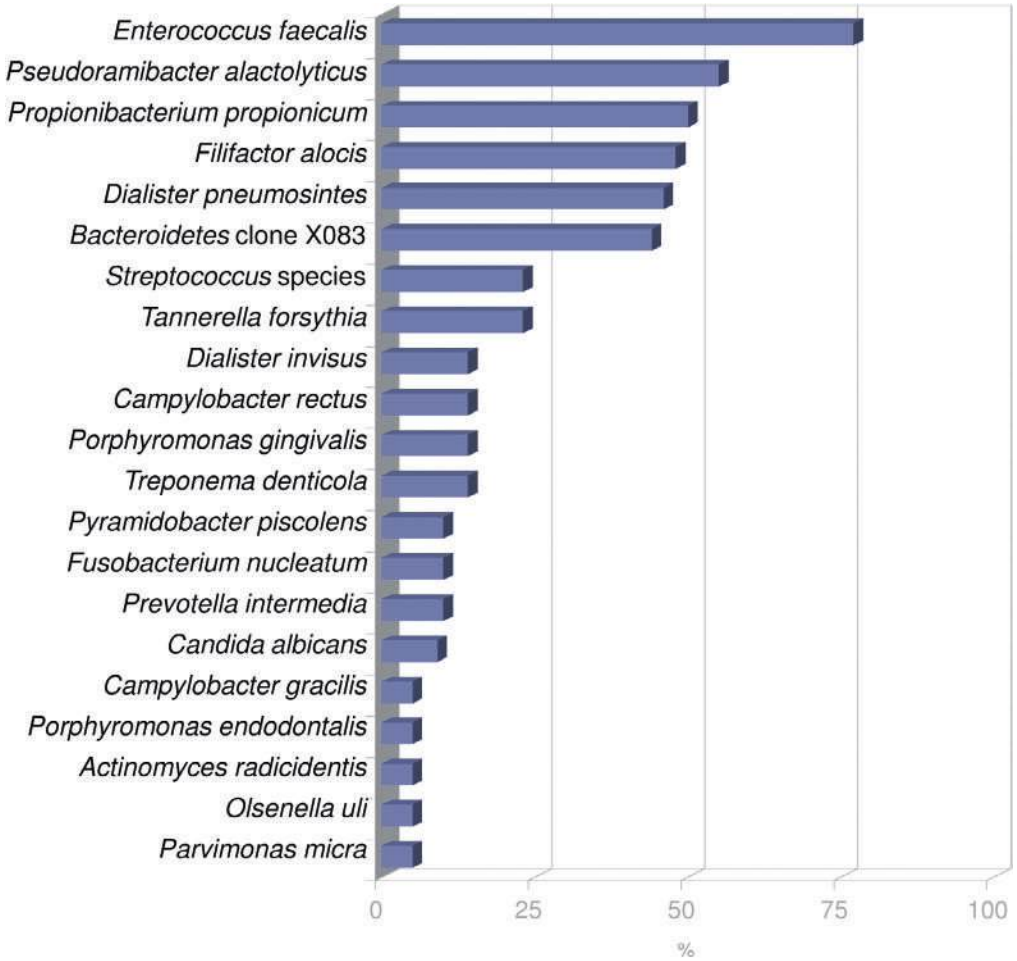


Fig. 15.28 Prevalence of microorganisms detected in root canal–treated teeth with posttreatment disease. Data from the authors' findings using a taxon-specific polymerase chain reaction assay.

chronic periradicular inflammation, sometimes resulting in an actively draining sinus tract.

- It can be a sequel to apical extrusion of debris during root canal instrumentation (particularly after overinstrumentation). Bacteria embedded in dentinal chips can be physically protected from the host defense cells and therefore can persist in the periradicular tissues and sustain periradicular inflammation. In addition, large bacterial flocs ("planktonic biofilms") may be located in the apical canal and be extruded during root canal procedures; once in the periapical tissues, they can induce or maintain inflammation (Fig. 15.32).[161] The virulence and quantity of the extruded bacteria, as well as the host's ability to deal with infection, will be decisive factors dictating whether an extraradicular infection will develop or not as a result of apical extrusion of debris.

Conceivably, the extraradicular infection can be dependent on or independent of the intraradicular infection.[215] Independent extraradicular infections are those that are no longer fostered by the intraradicular infection and can persist even after successful eradication of the latter. For an extraradicular infection to be classified as independent of the intraradicular infection, a comprehensive analysis of the apical canal should demonstrate absence of infection. So far, only a few cases have been reported that fulfill this requirement.[161,168]

It is still controversial whether asymptomatic apical periodontitis lesions can harbor bacteria for very long beyond initial tissue invasion.[12] Studies using culture-dependent[270,284,304] or culture-independent molecular biology methods, such as checkerboard hybridization,[59,272] FISH,[271] clone library analysis,[77] and HTS,[189] have reported the extraradicular occurrence of several bacterial species, especially anaerobes, in teeth that did not respond favorably to the root canal treatment. Because of the cross-sectional nature of the studies, it is not possible to infer if the extraradicular bacteria were transient or established therein. Apart from a discussion about whether contamination can be effectively prevented during surgical sampling of apical periodontitis lesions, these studies did not evaluate the bacteriologic conditions of the apical part of the root canal either. This makes it difficult to ascertain whether those extraradicular infections were dependent on or independent of an intraradicular infection. Actually, findings from a study evaluating both the resected root ends and the apical periodontitis lesions from treated teeth suggested that the large majority of cases of extraradicular infections are maintained by a concomitant apical root canal infection.[269]

Extraradicular biofilms are infrequent in untreated teeth with apical periodontitis, and when present they are virtually always associated with intraradicular biofilms; the occurrence of a sinus tract is usually indicative

of the presence of an extraradicular biofilm.[159,164] The high success rate of nonsurgical root canal treatment[36,162,259] suggests that, in the occasional cases it is present, the extraradicular infection component may not suffice to maintain apical periodontitis provided the intraradicular infection is properly managed. This is consistent with the concept of dependent infection. Even in root canal–treated teeth with recalcitrant lesions in which a higher incidence of extraradicular bacteria has been reported, a high rate of healing following retreatment[53,259] suggests that the major cause of posttreatment disease is located within the root canal system, characterizing a persistent or secondary intraradicular infection.[108,121,150,167,184,230,277,309,310]

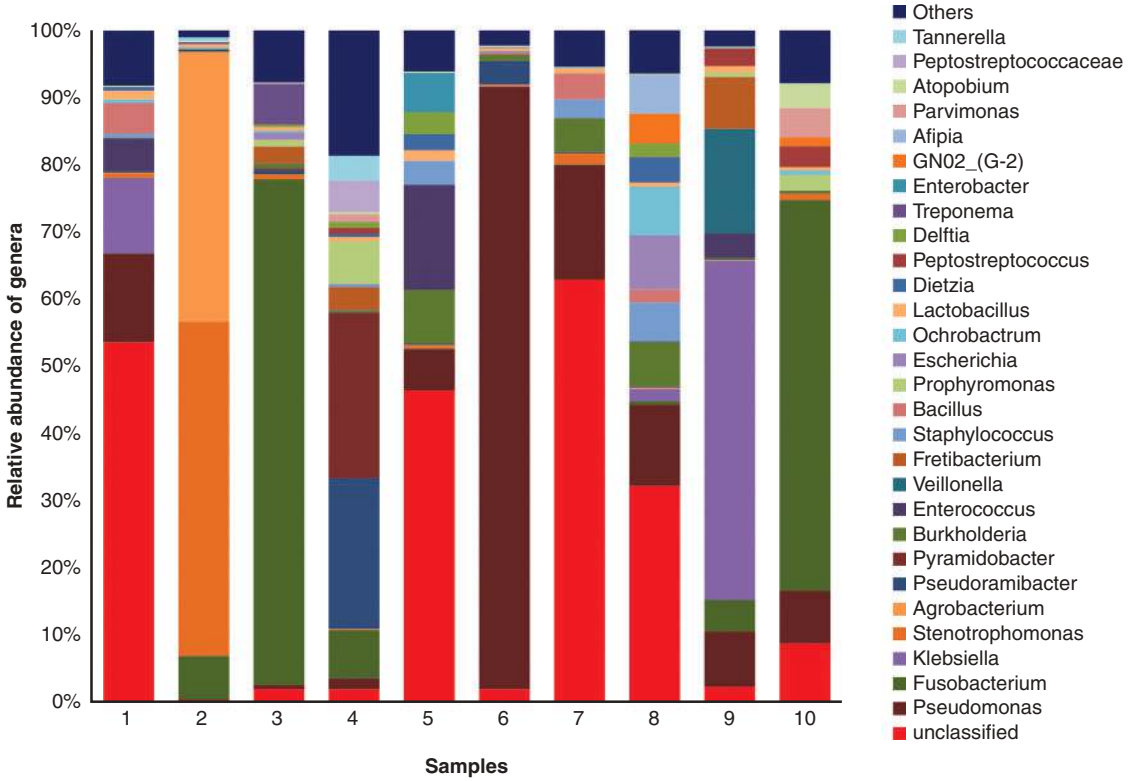


Fig. 15.29 Interindividual variability in bacterial diversity (richness and abundance) at the genus levels in cryopulverized apices from teeth with post-treatment apical periodontitis. Notice that no two samples are the same in terms of bacterial composition. (Based on data from Siqueira JF Jr, Antunes HS, Rôças IN, Rachid CT, Alves FR. Microbiome in the apical root canal system of teeth with post-treatment apical periodontitis. *PLoS One* 11:e0162887, 2016)

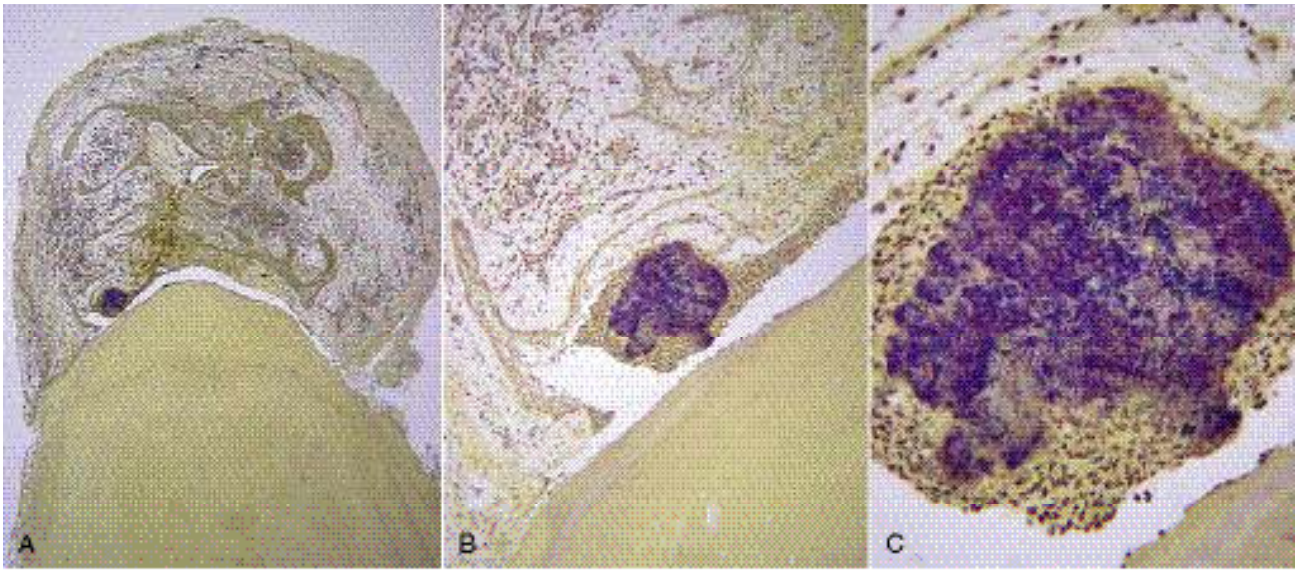


Fig. 15.30 Bacterial aggregate in an epithelialized apical periodontitis lesion, suggestive of actinomycosis. This case presented a severe flare-up in the past, but the tooth was asymptomatic at the time of extraction. **A,** Section not passing through the canal; note cyst cavity and bacterial aggregation on the left (magnification ×25). **B,** Detail of area with colony (magnification ×100). **C,** Higher magnification of colony, which is surrounded by neutrophils (magnification ×400). Sections stained with a Taylor-modified Brown and Brenn technique. (Courtesy Dr. Domenico Ricucci.)

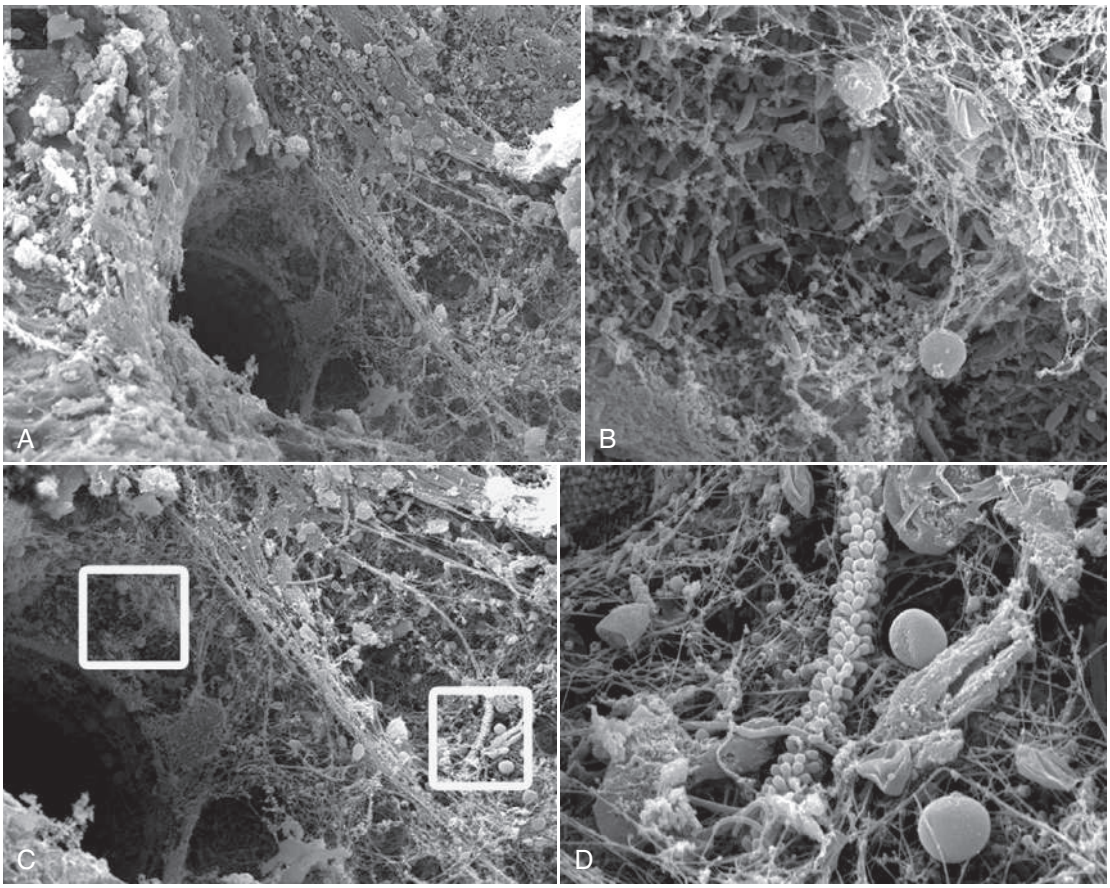


Fig. 15.31 Scanning electron micrograph showing extensive bacterial colonization in the very apical part of the canal, near and at the apical foramen (**A–B,** magnification ×550 and magnification ×850, respectively). **C,** Higher magnification of left inset in **B** showing a bacterial biofilm adhered to the very apical canal walls (magnification ×3700). **D,** Higher magnification of right inset in **B** showing a fully developed "corncob" in tissue meshwork adjacent to the apical foramen (magnification ×4000). (Modified from Siqueira JF Jr, Lopes HP: Bacteria on the apical root surfaces of untreated teeth with periradicular lesions: a scanning electron microscopy study. *Int Endod J* 34:216, 2001.)

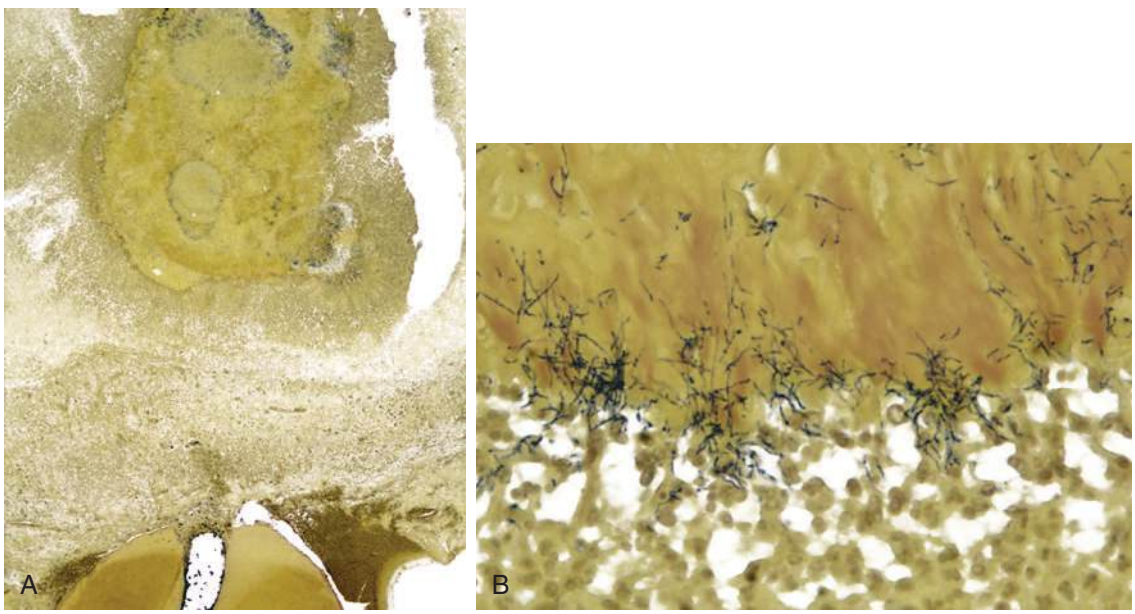


Fig. 15.32 A, Large bacterial floc located in the center of the apical periodontitis lesion, at some distance from the apical foramen. This represents an independent extraradicular infection that was the cause of persistent disease (magnification ×16). **B,** Higher magnification of the periphery of the large colony, showing intense accumulation of inflammatory cells (magnification ×400). (From Ricucci D, Lopes WP, Loghin S, Rocas IN, Siqueira JF Jr. Large bacterial floc causing an independent extraradicular infection and posttreatment apical periodontitis: a case report. *J Endod* 44:1308, 2018.)

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